Talk with your latent - can LLM do scientific reasoning directly from data?

Ilay Kamai[*]
Technion - Israel Institute of Technology
ilay.kamai@campus.technion.ac.il

Marc-Huertas Company
Instituto de Astrofísica de Canarias

Michael J. Smith
Center for Astrophysics, Harvard & Smithsonian

Alex M. Bronstein
Technion - Israel Institute of Technology
Institute of Science and Technology Austria (ISTA)

Hagai B. Perets
Technion - Israel Institute of Technology

Abstract

Recent advances in Large Language Models (LLMs) offer unique opportunities for scientific tasks, yet their ability to reason over complex numerical data remains largely unexplored. While several works investigate the ability of LLMs to create textual descriptions of scientific objects (e.g., proteins), we ask whether they can understand those objects and answer comparative questions about them. This kind of reasoning beyond description is crucial for the development of useful co-scientists. We test two different astrophysical datasets, involving both stars and galaxies, with pre-trained multimodal features that serve as the LLM data tokens. Adding those tokens as additional information, we fine-tune an 8B Llama model on various scientific description and reasoning tasks. We find that the model can describe the objects and compare pairs reasonably well in both datasets. On reasoning tasks, the outcome depends strongly on how the language model is adapted. Fully fine-tuned, the model can find cross-modal anomalies on galaxies, but it fails to identify outliers within a population of stars, and its performance drops with the size of the population. Training only a low-rank adapter (LoRA) on a frozen base model, with neither method's hyperparameters tuned, nearly doubles the stellar score, removes the dependence on population size, and improves the galaxy task; most of the gain remains when the two methods share one optimizer schedule. We further show that a small flow-matching model, which receives catalog labels as auxiliary data and returns its output to the LLM as latent tokens rather than as text, improves the identification of outliers that were never outliers in training ($0.62 \rightarrow 0.75$). We call this *latent tool use (LTU)*. Frontier general-purpose models given the raw observations zero-shot, without tools, score well below these narrators. In addition, we show that for such questions, a template-based dataset is preferable to an LLM-based dataset. The results indicate that a language model reading learned scientific features can perform population-level reasoning when it is adapted with a low-rank adapter, and that a dedicated model given auxiliary labels adds to this where the training data offers no precedent.

[*]Corresponding author.

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1 Introduction

The integration of Large Language Models (LLMs) into the scientific workflow marks a paradigm shift in how researchers interface with information. While these models have demonstrated remarkable proficiency in some scientific tasks, their utility remains largely confined to the textual and visual domains. A critical frontier in "AI for Science" is the transition from processing scientific text to reasoning directly over the physical data that defines scientific discovery. Standard LLM architectures struggle with numerical reasoning [1], in part because subword tokenization does not preserve the magnitude and continuity of numerical values [2], creating a gap between natural language and empirical data.

The main motivation of this work is the need to combine physics foundation models and LLMs in a scientifically useful manner. Each has distinct advantages, and the potential for knowledge alignment between them has far-reaching outcomes. One is interpretability: an LLM that operates over both natural language and learned scientific representations could serve as a 'bridge' between humans and machine learning representations, which are usually hard to interpret. A second is scientific discovery: LLMs have broad theoretical knowledge, and combining this general knowledge with specific data can lead to new discoveries. In addition, physics foundation models are trained on large and diverse datasets and encapsulate latent information beyond their downstream tasks, and that information can support non-trivial population analysis or complex cross-modal correlations. Unfolding it, however, requires analysing high-dimensional features directly, whereas the common practice of projecting them to a low-dimensional summary discards exactly the structure such analyses depend on. Most current uses of LLMs with scientific data take the form of textual descriptions, or of agents that delegate the analysis to dedicated models and read off their results as text, which is usually called *tool use* [3–5]. Whether an LLM can perform scientific reasoning *directly* from learned data representations is therefore still an open question, and it is the question we study here. Where it cannot, we ask a second one: whether the result of a dedicated model can be handed to the LLM in a form it can use, as latent tokens rather than as text. We call this *latent tool use (LTU)*.

Contribution.

First, we show that the way the language model is fine-tuned decides whether it can reason over a group of objects. With full fine-tuning, the model describes single objects and compares pairs well, but it often fails to find the odd star in a group, and it fails more often as the group grows. With LoRA on a frozen base, the same model, trained on the same data, finds the odd star in most groups at every group size, and it also judges galaxies more accurately. A linear probe shows that the features contain the answer, so the fully fine-tuned model fails because it does not read them well.

Second, we introduce latent tool use. A small model compares each star with the rest of its group, using the star's features and catalog labels, and returns its result to the language model as one latent token per star. The language model uses this token without copying the small model's answer, and answers more accurately on groups unlike those it saw in training.

Third, we show that training on verified templated prose gives a more factual model than training on fluent LLM-written prose, because LLM-written prose states more than its input supports.

2 Related Work

LLMs as scientific assistants. Advances in LLMs have already had a wide-ranging impact on how we do science. LLM agents can now review and write scientific papers [6, 7], act as a 'virtual lab' that suggests and implements novel research ideas [8], or perform equation discovery [9]. However, these agents typically access scientific data indirectly, by writing code to an external tool (e.g., ESM, AlphaFold-multimer), an approach formalised in the tool-augmentation literature [3, 4]. Recently there has been growing interest in predicting physical properties of proteins with LLMs [10, 11]. That line is close to ours, but asks a different question: property prediction and textual description from sequential data, rather than comparative and population analysis over continuous data.

Scientific foundation models and scientific LLMs. Foundation models have become popular across scientific fields and show strong performance in many domains [e.g., 12–19]. Notably, these models do not treat natural language as a modality, so a clear separation persists between scientific

foundation models and LLMs. The term scientific-LLM usually refers instead to injecting scientific knowledge into existing LLMs [for reviews, see 20, 21], most often by fine-tuning on high-quality text and images to improve reasoning [22–24], object description [25], search [26], and chat [27]. Such models outperform general LLMs on domain tasks [28], but remain confined to text and images.

Connecting frozen encoders to frozen language models. The architecture we use is standard outside the sciences: a trained projector maps a frozen encoder's outputs into the embedding space of a frozen language model, which is adapted with a small number of trainable parameters [29–32]. This recipe is well-established for producing descriptions of a single input. Whether it supports comparison across several inputs, which is what population-level scientific questions require, has received little attention.

Tool use, in text and in latent space. In tool-augmented language models the model decides when to call a tool, writes its arguments, and reads the result back as text [3, 4]; ToolkenGPT makes the call itself a learned token embedding, but the result still returns as text [5]. Closer to our setting, Prismer feeds the outputs of expert vision models to a language model as tokens [33], and PEARL trains a vision-language model to predict the effect of a tool in latent space, so that no tool is called at inference [34]. Latent tool use, as we use the term, differs from both: the tool is run, on every input, and its result is returned to the language model as latent tokens that the model is trained to read, instead of as text. The language model does not decide when to call it; we leave that to future work.

3 Data and Model

Terminology. We use six names throughout. The *narrator* is the 8B language model we train: it reads an object's data tokens and answers questions about the object. The *tool* is a small model, not a language model, trained separately to compute, for every object in a question, a quantity that the question depends on (Section 4). It is run on every input, and its output enters the narrator as one extra token per object, the *tool token*, which the narrator is trained to read; a narrator that reads it is a *tool narrator*. The stellar tool also receives catalog labels as inputs (Section 4), and narrators without a tool do not. The *templater* is the deterministic program that writes our training text from catalog facts, and the *writer* (DeepSeek-V4) is the LLM we use in its place in one comparison. Generated text is scored by the *verifier*, a program that checks every stated fact against the catalog, and by the *judge* (Qwen3.5-27B), an LLM that reads each description against the ground truth.

We test two different astrophysical datasets, representing different domains and physical properties. The first dataset is a *stellar* multimodal dataset comprising 190,091 stars, observed by 5 different modalities coming from different telescopes - *Kepler*[35], TESS[36], LAMOST[37], APOGEE[38], and Gaia[39]. Each modality is processed through a specific encoder, and the modalities are aligned with a model that factorizes the latent space into shared and specific subspaces. The result is one shared feature and 5 per-modality features, each projected to the token embedding space of the LLM as its own small group of tokens. The second dataset is galaxies from the COSMOS-Web survey [40], including two modalities: images and star-formation history (SFH). Similar to the stellar dataset, we use a pretrained model that aligns the two modalities (here using CLIP [41]) and project the frozen features into the LLM's token embedding. Notably, the two datasets are different in many aspects: number and type of modalities, pretrained models, scientific questions, and what their test sets hold out (below). Using the two datasets in the same pipeline highlights general vs data-specific conclusions.

training prose. One of the challenges in this pipeline is creating the training dataset. This should include ground-truth prose per object. Contrary to other fields, in astrophysics data is usually stored only in numerical catalogs, so one needs to create the prose from scratch. Our main method is a *template-based* method: a deterministic program that picks a subset of facts from the catalog, chooses a phrasing from a bank of templates, and writes prose. Every sentence is checked against the record by the verifier before it enters the corpus. The check is per-fact: it guarantees that the facts a description states are true of the object, not that the description mentions every unusual object in a group. Both datasets share two types of prose: description of a single object (single), and comparison between pairs (pair). We also test deeper reasoning capabilities with two additional tasks, and this is where the datasets diverge. In the stellar dataset, we use population prose: a group

of N objects with a description of the typical members, and a callout of the ones that do not belong and on which property. In the galaxy dataset, the task is to judge whether a galaxy's shape agrees with its star-formation history, a relation between its two modalities.

Objects, facts and descriptions. The two datasets are split in different ways, and the difference decides what each test measures. The galaxy set has $137{,}696$ galaxies, split $80/10/10$ into training, validation and test galaxies, and all galaxy training text is written about training galaxies only: the galaxy test holds out the objects themselves. The stellar set has $190{,}091$ stars, of which we evaluate on a held-out set of $19{,}010$, but the stellar training text was written for all stars and divided into training and validation by description, not by star. As a result, 99.2% of the evaluation stars appear in *some* training description, and the stellar test holds out questions rather than objects. This matches how a stellar catalog is used: the stars of the Kepler field are fixed and already catalogued, and the questions of interest are new comparisons among them, not descriptions of stars never observed. No test group appears in training (the largest overlap between a test group and any training group is a third of their combined members), and in Section 4.4 we also hold out the answer itself. For questions about a single star the overlap does matter, because the narrator can reuse what training said about the same star. Section 4.4 measures how much, with the galaxies, where no such overlap exists, as the comparison.

Each star's catalog record holds 13 numbers ($T_{\rm eff}$, $\log g$, $[{\rm Fe/H}]$, mass, radius, age, luminosity, distance, parallax, Gaia's astrometric quality index RUWE, apparent and absolute brightness, and rotation period) and categorical facts: evolutionary state (dwarf, subgiant, giant); how the star compares with the rest of the sample (e.g., among the most metal-rich quarter, or hotter, older or faster-rotating than most); its Galactic disk population and α-element abundance; its activity level; and flags for binarity, planet hosts, flares and open-cluster membership. Each galaxy's record holds stellar mass, star-formation rate and state (quiescent to starburst), redshift, Sérsic index (how centrally concentrated its light is), visual morphology and disturbance, formation age, and the shape of its star-formation history (rising or falling, recent activity, quenching, formation time).

The templater writes each description by sampling, independently, which facts to state (a random three to six), a style, and a framing. For stars it also adds, each with probability one half, a chemistry block (disk population, α-abundance) and an activity block (chromospheric activity, variability amplitude). Stars have three styles (natural prose, observation log, bullet list) and four framings (plain description, outlier flag, comparison with other stars, comparison with the Sun). Galaxies have four styles (technical paper, observation log, teaching text, concise summary), four framings (plain description, comparison with the galaxy population, comparison with the star-forming main sequence, outlier flag) and three lengths. On stars, each fact is then worded by drawing from a bank of alternative phrasings (917 alternatives at 141 points in the text); the galaxy templater has one wording per fact and style, so its variety comes from the facts, style, framing and length. A star is described about five times, each time with different facts and words (Figure 1). The result is highly diverse: every stellar single and pair description is unique, and 99.9% remain unique with all numbers masked; for galaxies, 99.9% of single descriptions are unique and 70% remain so with numbers masked. In total, the stellar corpus has 1.93M records (1.01M single, 0.69M pair, 0.23M population); each population record is a group of 4 to 32 stars with an odd member on one of six properties. The galaxy corpus has 288k records (90k single, 198k pair). The galaxy agreement task uses a separate, deliberately narrow corpus of $45{,}817$ records (three phrasings per case, also from training galaxies only) in which galaxies whose morphology and star-formation history disagree are oversampled to 33%.

Training recipe. A narrator trained on templated text can learn to reproduce the templates instead of reading the data. We counter this with two changes to the input. First, we vary its format: an object's data tokens appear alone, after a question or a short passage, separated from the description by intervening text, or among the tokens of several other objects together with a request to describe one of them. Second, part of the intervening text consists of real sentences from astrophysics papers, filtered so that none states a relation between properties we later test. The two are needed together (Section A.3). Varying the format alone improves accuracy but makes the narrator copy training descriptions almost verbatim; adding the paper sentences removes that copying, and accuracy improves further. Population records are always presented as the data tokens alone. The recipe was chosen on the galaxy data and reused unchanged for stars.

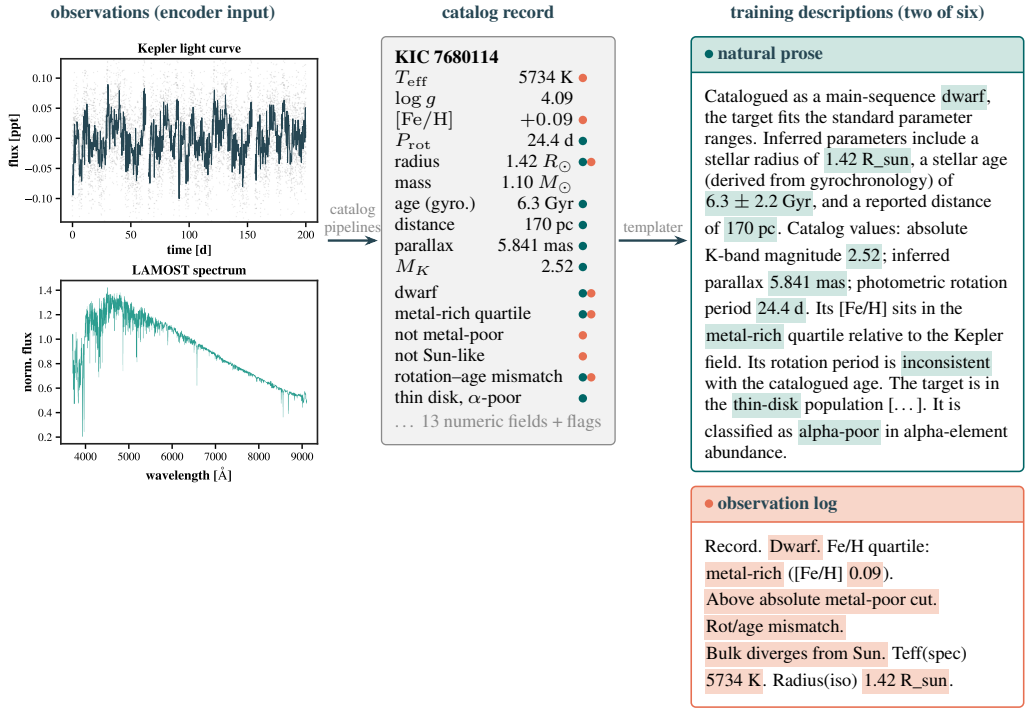


Figure 1: **From observations to training text**, for one star (KIC 7680114). *Left*: two of its observations, the Kepler light curve (6-hour bins over 30-minute data) and the LAMOST spectrum. The frozen encoders read these; the language model never sees them directly. *Middle*: its catalog record (abridged). *Right*: two of the six training descriptions written about it, verbatim, in two styles. Each draws a different random subset of facts. Dots mark which facts each description uses.

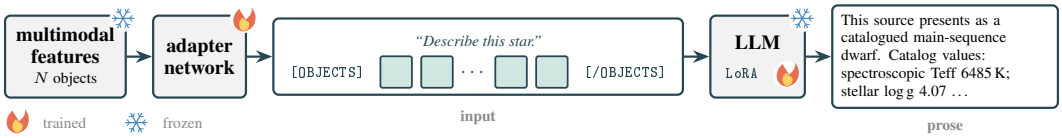


Figure 2: **The pipeline.** An adapter projects each object's multimodal features into tokens; the model processes them, outputs prose, and the loss is taken on the prose alone. The language model is adapted either by full fine-tuning or with LoRA.

4 Results

4.1 singles and pair questions

We first test the single and pair questions, comparing two types of Llama-8B models: Llama-3.1 and AstroSage [28], a Llama-3.1 model that was fine-tuned specifically on arXiv astrophysics papers. Table 1 shows the comparison on both datasets. Single and pair descriptions are scored by the verifier, which extracts every numeric and categorical claim from the generated text and checks it against the catalog: *numeric accuracy* is the fraction of stated values within a per-quantity tolerance of the catalog value, and *qualitative* (single) or *relational* (pair) accuracy is the fraction of class words or comparisons that match it (tolerances and definitions in Section A.4). Both models perform well on both tasks, and the differences between them are split evenly: AstroSage is ahead in four of the eight cells, and no difference exceeds 0.05. We use AstroSage for the rest of the paper. Section A.5 breaks the accuracy down by property: class statements are mostly right, while some stated numbers (age, parallax, galaxy stellar mass) are often off by more than 10%. We also trained linear probes on the same tokens (Section A.7). On galaxies, the narrator states about as much as the probes recover. On stars, the narrator also recalls some catalog values from the training text, so part

of its numeric accuracy is memorized. Where we can compare them directly, the probes are more accurate. However, each probe predicts one property on one dataset, while a single narrator answers about every property on both datasets, in free text.

4.2 scientific reasoning tasks

Single and pair questions consists of description and simple comparisons and requires almost no reasoning beyond the basic properties. It is also similar to the protein description tasks as in [11], and therefore it is not surprising that an LLM performs well on such questions. Next, we test the performance on questions that requires some form of scientific reasoning. As mentioned in Section 3, those are finding the odd member of a group of stars, and judging whether a galaxy's shape agrees with its star-formation history. The rest of the paper focuses on those tasks. Table 2 shows the results of the two reasoning tasks. Figure 4 shows one example generated description for each task. On both tasks, the result depends on how the language model is fine-tuned. On stars, the fully fine-tuned narrator reads the data tokens and fails: it names the odd member and its property in only a third of the groups. The same model trained with LoRA on a frozen base, on the same text and data tokens, does so in 63% of groups. On galaxies, the LoRA narrator gives the right verdict for 97–99% of galaxies, and the fully fine-tuned narrator is worse, especially on quiescent galaxies (0.88 against 0.97). We did not tune the hyperparameters of either method, and the two methods used different optimizer settings. We therefore trained a second stellar LoRA narrator with the optimizer settings of the fully fine-tuned one. It still scores well above full fine-tuning (0.51 against 0.33; Section 4.3).

A cross-fit linear probe on the same aligned features, asked to rank the members of a group, matches what the true catalog values achieve (rank-1 0.660 from the shared subspace and 0.686 from the full feature, against 0.660 for the catalog itself; Section A), which implies that the signal the task needs is present in the features the model reads, so the failure is in the readout and not in the representation.This aligns with the previous observation that LoRA and fully fine-tuned models get the same features and perform very differently. The last column of Table 2 adds the tool token (see Section 3). In the stellar case the tool is a flow-matching model trained on the joint distribution of a star's aligned features and its catalog values ($T_{\rm eff}$, $\log g$, $[{\rm Fe/H}]$, rotation period, binarity and cluster membership), conditioned on the rest of the group. For each star it returns how surprising the star is under that model and the values the group would predict for it, eleven numbers in all, which a trained projection maps to that star's tool token. The catalog values are therefore inputs of the tool at test time as well as in training. This is inherent to how it works, since its surprise score is conditioned on them. Narrators without the tool see only the aligned features. In the galaxy case, it is a cross-modal predictor trained to predict one modality from the other, and it returns one token per galaxy. The stellar tool is not an answer on its own: ranking the members by its surprise score puts the odd star first in 29% of groups. On stars, the tool narrator doubles the score of the fully fine-tuned narrator ($0.33 \rightarrow 0.66$) and names the odd star in 80% of groups, but most of that difference comes from the training: against the LoRA narrator without the tool, the tool adds 0.03 ($p = 0.07$; Section 4.3). When the tool's own top choice is wrong, the tool narrator follows it no more often than the fully fine-tuned narrator does (7% against 8%). The narrator uses what the tool returns; it does not copy a verdict from it. The gain over the fully fine-tuned narrator holds for every kind of odd star, and also on the 556 groups in which the odd star is the only member with its odd property ($0.37 \rightarrow 0.71$, with 0.66 for the LoRA narrator without the tool; Section A.6). On galaxies, it is no better than a narrator trained the same way without the token (Section A.2).

For the fully fine-tuned narrator, the asymmetry between the two tasks, together with the results in Table 1, points to the number of objects in the question as the important factor for successful generation of the *narrator*. The galaxy verdict concerns a single object and follows from two of its own properties, its shape and whether it still forms stars, which the narrator can read from that galaxy's tokens. Even though morphology-SFH disagreement is deeper than mere description, the *narrator* still performs almost perfectly. In the stellar case, finding the odd star requires comparing every member with the rest of the group, a computation over the whole set that no single star's tokens contain. That narrator's strong performance on single and pair questions supports the hypothesis that population-related questions are harder for it. To test this, we vary only the number of stars while keeping the question fixed: the test groups contain between 4 and 32 stars, and Figure 3 shows the score as a function of group size. With full fine-tuning, the score falls steadily as the group grows, from 0.45 with 4 stars to 0.06 with 32 (Spearman $\rho = -0.24$, $p = 10^{-16}$). The LoRA narrator without the tool stays between 0.53 and 0.70 with no significant trend ($\rho = +0.01$, $p = 0.7$), and

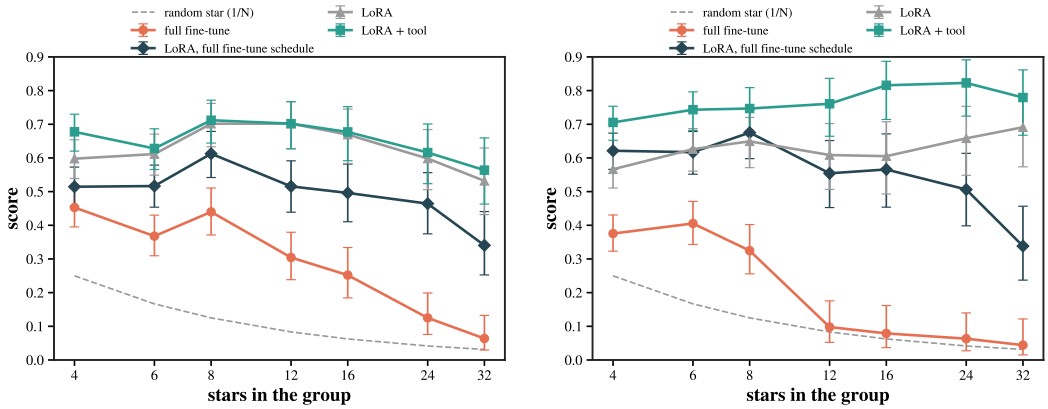


Figure 3: **Only the fully fine-tuned narrator degrades with the number of stars.** Score of Table 2 (the narrator names the odd star and its property) as a function of the number of stars in the group. Left: the 1,203 test groups with an odd member. Right: the 1,000 groups whose odd member was never odd in training (Table 7). Error bars are 95% intervals. Dashed line: the chance of picking the odd star at random ($1/N$); the score also requires the right property, so its chance level is lower still.

Galaxy: does its shape agree with its history?

Catalog. Late-type disk (P=0.75); quiescent ($\log \mathrm{sSFR} = -11.6$); declining star-formation history. A disk that has stopped forming stars: shape and history *disagree*.

Narrator (no tool token) ✓

A quiescent late-type disk — a cross-modal outlier, because disks rarely quench while retaining their morphology. Its star-formation history is declining.

Shape, star formation and verdict all match the catalog.

Stars: which one does not belong?

star	class	T_{eff} [K]	$\log g$	R [$R_\odot$]
0	dwarf	5778	4.39	1.05
1	**giant**	4662	**2.45**	**11.14**
2	subgiant	5628	3.89	2.00
3	dwarf	5394	4.55	0.81

Narrator, no tool ×

These 4 stars form a homogeneous population of field stars. No anomalies detected along the field-star axis.

Narrator + tool ✓

Sample N=4: single stars. index 1: a giant [logg = 2.44 dex, R = 11.14 Rsun, logL = 1.72 log(Lsun), giant].

Figure 4: **Example generations on the two reasoning tasks**, verbatim and decoded greedily. *Left*: a quenched disk, whose shape and star-formation history disagree. The narrator trained without the tool token states the shape, the star-formation state and the verdict correctly, as it does for 97–99% of test galaxies. *Right*: a test group of four stars containing one giant. The fully fine-tuned narrator without the tool reports no odd member; the LoRA narrator with the tool names the giant, with values that match the catalog.

the tool narrator between 0.56 and 0.71 ($\rho = -0.03$, $p = 0.3$). With the optimizer schedule of the fully fine-tuned narrator, the LoRA narrator declines only mildly, from 0.51 to 0.34 ($\rho = -0.06$, $p = 0.04$). The same pattern holds on groups whose odd star was never odd in training (Figure 3, right). The loss of accuracy with group size is therefore a property of the fully fine-tuned narrator, not of a language model reading a group. Since every star adds the same number of tokens, a larger group is also a longer input, and our data cannot separate the two. Next, we separate the training from the tool, and then test whether the improvements come from reasoning or memorization and whether handing a model the full catalog numbers prevents the failure.

4.3 is it the token, or the training that came with it?

The narrators with and without the tool are separately trained models, and the fully fine-tuned narrator differs from the tool narrator in more than the token: the tool narrator is trained with LoRA

Table 1: **Base model: Llama-3.1-8B vs. AstroSage-8B**, identical corpus, recipe and seed; rule-scored, greedy; the better model in bold. Metrics: *numeric acc.* = fraction of numbers the text states that are within 10% of the catalog value (Section A.4); *qualitative acc.* (single) = fraction of class words (dwarf/giant, disk/elliptical, ...) that match the catalog; *relational acc.* (pair) = fraction of comparisons ("hotter than", "more massive") that match.

metric	Llama-3.1-8B	AstroSage-8B
galaxies — one object		
numeric acc.	**0.750**	0.711
qualitative acc.	0.881	**0.896**
galaxies — pair		
numeric acc.	0.410	**0.444**
relational acc.	**0.795**	0.771
stars — one object		
numeric acc.	0.651	**0.660**
qualitative acc.	0.888	**0.891**
stars — pair		
numeric acc.	**0.776**	0.766
relational acc.	**0.942**	0.900

Table 2: **The reasoning tasks.** Each row is one task and scorer; the columns are narrators that differ in how they were trained and whether they read the tool token. Stars: the fraction of the $1{,}203$ test groups with an odd member in which the text names it *and* its property, scored by the verifier and by the judge. Galaxies: the fraction of test galaxies with the correct agreement verdict (Section A.4), on the $9{,}609$ star-forming and $1{,}068$ quiescent galaxies.

		narrator, no tool token		narrator + tool
task	scorer	full fine-tune	LoRA	LoRA
stars — name the odd member	verifier	0.245	0.564	**0.633**
and its property	judge	0.332	0.633	**0.662**
galaxies — shape vs. history,	star-forming	0.980	**0.988**	0.986
correct verdict	quiescent	0.884	0.974	**0.978**

on a frozen base and with a different optimizer schedule. To separate these factors we trained two further stellar narrators without a tool (Table 3): one with exactly the recipe of the tool narrator, and one with LoRA but the optimizer schedule of the fully fine-tuned narrator (Section A.9). All four narrators share the training text, prompt, base model and test items, and each column of Table 3 changes one thing from the previous one.

LoRA instead of full fine-tuning. With the same optimizer schedule, LoRA raises the score from 0.33 to 0.51 on the test groups ($p < 10^{-24}$) and from 0.28 to 0.59 on the groups whose odd star was never odd in training ($p < 10^{-60}$). This run had not converged when it was stopped. It finds the odd star more often but makes more contradictions than the fully fine-tuned narrator (2.32 against 1.90 per group), so what LoRA brings at this schedule is the identification of the odd star, not a more accurate description overall.

The optimizer schedule and training length. Moving to the schedule of the tool narrator adds 0.12 on the test groups ($p < 10^{-12}$), only 0.03 on the never-odd groups ($p = 0.14$, not significant), and removes about two thirds of a contradiction per group on both sets ($p < 10^{-37}$). Since the previous run had not converged, we attribute this step to the schedule and the training length together. Neither recipe was tuned, so a better recipe for full fine-tuning may exist; LoRA is better without one.

The tool. With the training held fixed, the tool adds 0.03 on the test groups ($p = 0.07$), and 0.13 on the groups whose odd star was never odd in training ($0.62 \to 0.75$, $p = 5 \times 10^{-15}$). Part of this gain lies in the groups whose odd star is a binary, 54% of this set ($0.48 \to 0.64$), and binarity is one of the catalog flags the tool receives. The gain remains without them ($0.78 \to 0.88$, $n = 458$, $p = 6 \times 10^{-6}$). Temperature and surface gravity are also inputs of the tool, as described above, so we read this result as follows: when catalog labels are given to the tool as auxiliary data, the narrator

Table 3: **Separating the training from the tool**, stellar group task, judge-scored, greedy, paired. *score*, *named* and *property* are as in Table 7; *covered*, *contradictions* and *all correct* as in Table 10. *Top*: the 1,203 test groups with an odd member. *Bottom*: the 1,000 groups whose odd member was never odd in training.

	full fine-tune	LoRA, FT schedule	LoRA	LoRA + tool
test groups (n = 1,203)				
score	0.332	0.510	0.633	**0.662**
named	0.442	0.672	0.766	**0.797**
property	0.391	0.556	0.660	**0.687**
covered	0.569	0.638	**0.658**	0.623
all correct	0.126	0.112	**0.226**	0.221
contradictions/group	1.90	2.32	**1.55**	1.57
groups whose odd member was never odd in training (n = 1,000)				
score	0.279	0.590	0.615	**0.748**
named	0.329	0.652	0.655	**0.782**
property	0.343	0.669	0.646	**0.761**
covered	0.560	0.586	0.627	**0.628**
all correct	0.017	0.023	0.104	**0.140**
contradictions/group	1.85	2.10	1.44	**1.35**

uses the returned token to find outliers for which the training data offers no precedent. We do not separate the contribution of the learned density from that of the labels, which the tool models jointly.

Table 4: **Breaking the token at test time**, stellar group task, full test set (n=1,203 groups with an odd member, greedy, judge-scored). *score* is the fraction of descriptions that name the true odd member and its property. The last three rows are one trained model, the tool narrator, with its token intact or damaged at generation time; the first two rows are separately trained models without a tool token.

	score
narrator, full fine-tune, no tool token	0.332
narrator, LoRA, no tool token	0.633
tool narrator, token zeroed	0.561
tool narrator, token scrambled	0.606
tool narrator, token intact	**0.662**

The tool narrator does read its token. At test time, on the same trained model (Table 4), scrambling the token across the group costs 0.056 and replacing it with zeros costs 0.101, both significant.

Galaxies and single objects. On galaxies we find the same ordering of the training methods (Section A.2): the LoRA narrator without a tool matches the tool narrator on quiescent galaxies (0.974 against 0.978, not significant) and is slightly better on star-forming ones, while the fully fine-tuned narrator gives the wrong verdict for 12% of quiescent galaxies. The galaxy tool adds nothing, as expected, since it is computed from the same image and star-formation embeddings the narrator reads. This galaxy comparison also differs in the optimizer schedule. On single stars and pairs (Table 5) the differences are smaller: the LoRA narrator with the schedule of the tool narrator is more accurate than the fully fine-tuned one, while with the full fine-tuning schedule it is equal on class statements and comparisons and lower on stated numbers. The advantage of LoRA over full fine-tuning at a matched schedule therefore appears in the group task.

4.4 what does the narrator memorize?

A narrator trained on templated text, and on the stars it is later tested on (Section 3), could copy instead of read. Three things could be copied: the wording of the training descriptions, the values written about a particular object, and the answer to a reasoning question. We test each where it can be tested (Table 6). The tests use different data, so each is read against its own control, not against the other rows.

Table 5: **Single stars and pairs, by training**, whole test set (19,010 single stars and 9,505 pairs), greedy, scored by the verifier as in Table 1. The tool token enters group questions only, so the last two columns are separately trained models that read the same inputs here.

	full fine-tune	LoRA, FT schedule	LoRA	LoRA + tool
single star, numeric acc.	0.666	0.615	0.724	**0.796**
single star, qualitative acc.	0.890	0.890	0.910	**0.933**
pair, numeric acc.	0.769	0.724	0.802	**0.849**
pair, relational acc.	0.939	0.947	0.953	**0.969**

Table 6: **What the narrator copies from its training text.** Each row is a separate test with its own control; numbers are not comparable across rows.

what could be copied	data	test	finding
wording	galaxies	training recipe (Section A.3)	copied when only the input format varies; the paper sentences remove it
values about one object	stars, galaxies	narrator against probes and against the training text (Section A.7)	exact values copied when the training text gives them; average error barely changes
the answer to a reasoning question	stars	test set split by whether the odd star was ever the odd one in training, and a larger set built that way (Table 7)	not recalled; LoRA's gain holds, and the tool's is larger

Wording. With a varied input format but no paper sentences, the narrator recites its training descriptions: the fraction of novel descriptions falls from 0.40 to 0.18 on single objects and from 0.42 to 0.04 on pairs. Adding the paper sentences restores it (0.39 and 0.68). Every narrator in this paper uses both.

Values about one object. On galaxies, where no test galaxy is described in training, the narrator states about what a non-linear probe on the same tokens can read. On stars it states more: it is more accurate than the same probe on most numbers (radius error 0.40 against $0.82\,R_\odot$), and it gives the exact temperature (zero Kelvin error) in 35% of its statements, against 0.5% for the probe. Almost every such exact value is written in some training description of the same star; where none gives it, exact matches fall to the probe's level. The average error changes little. Part of the stellar narrator's numeric precision is therefore recall, and its single-star numbers in Table 1 should be read with this in mind.

The answer. The objection that matters most is that a narrator might recall which star was called unusual during training rather than work it out. The comparisons in Table 2 are fair, since every narrator saw the same training text, but they do not rule this out. We test it twice. First on the test set itself: we split its $1{,}203$ groups with an odd member by whether that member was ever the odd member of a training group. The two halves are disjoint and together give the numbers of Table 2 exactly (Table 7, top). Almost every odd star was odd somewhere in training ($1{,}167$ groups), as the overlap described in Section 3 implies, and 36 groups remain whose answer never appeared. On those the tool narrator gains 0.31 over the fully fine-tuned narrator ($0.39 \to 0.69$; $p = 0.003$, bootstrap 95% interval $+0.14$ to $+0.47$), against 0.33 on the other half ($0.33 \to 0.66$), and the LoRA narrator without the tool reaches 0.56 ($p = 0.07$ against full fine-tuning, with only 36 groups). Second, because that half is small, we build a larger set of the same kind: $1{,}000$ new groups from the test fold, each with an odd member that was never an odd member in training, which keeps 2.3% of candidate groups. Table 7 (bottom) gives the answer with more power. LoRA alone gains 0.34 over full fine-tuning, so that gain is not recall. The tool adds 0.13 on top, compared with 0.03 on the test groups. This set is harder by construction, since the stars that survive the filter are rarely odd and their anomalies are subtler, so its absolute rates are not comparable to those of Table 2; only the gap within it is.

Taken together, the narrator copies surface detail, namely wording when the training allows it and exact values it has read about the same star, but not conclusions: whether a star is the odd one in its group is worked out from the input, not recalled. This is also why the stellar overlap does not

Table 7: **Is the answer recalled?** Judge-scored. *Top*: the test groups of Table 2, split by whether the odd member was ever the odd member of a training group; the two rows are disjoint and together reproduce Table 2. *Bottom*: a larger set of 1,000 new groups whose odd member was never odd in training. Δ is the tool narrator against the LoRA narrator. *score* is the fraction of descriptions that name the true odd member and its property; *named* counts naming the odd member regardless of property, and *property* naming the right property regardless of member.

		groups	full fine-tune	LoRA	narrator + tool	Δ
test set, odd star was odd in training	score	1,167	0.330	0.636	**0.661**	+0.025
	named		0.441	0.771	**0.800**	+0.029
	property		0.388	0.661	**0.686**	+0.025
test set, odd star never odd in training	score	36	0.389	0.556	**0.694**	+0.138
	named		0.472	0.583	**0.694**	+0.111
	property		0.472	0.639	**0.750**	+0.111
new groups, odd star never odd in training	score	1,000	0.279	0.615	**0.748**	+0.133
	named		0.329	0.655	**0.782**	+0.127
	property		0.343	0.646	**0.761**	+0.115

undermine the reasoning results. They are measured on new groups, the gain from LoRA is as large where the answer never appeared in training, and the tool's gain is larger there.

4.5 giving the model the numbers as text

If the features carry the signal and the model cannot use it, an obvious alternative is to skip the features and hand the model the catalog numbers directly. We test this on the stellar task. Each group is shown as a table with one row per star, listing its catalog values (temperature, surface gravity, metallicity, radius, mass, and others) together with the two catalog flags the tool also receives: whether the star is a binary and whether it belongs to an open cluster. With these flags the table holds everything the tool is given, so the comparison is fair for all five kinds of odd star. The model sees two worked examples and is asked, in words, to name the star that does not belong and the property it differs in. We test two models: the base model (AstroSage-8B), without our training, and Qwen3.5-27B, a larger general-purpose model. Each is tested twice: answering directly, and reasoning step by step before answering. Table 8 compares them with the two narrators on the same 1,203 groups.

No model that reads the numbers reaches the LoRA narrators (0.633 without the tool and 0.662 with it; $p < 10^{-14}$ for the best text reader against the LoRA narrator without the tool). AstroSage reading the table reaches the best score among them (0.473), but by naming most of the group (82% of the stars): naming the odd star among them is then no better than chance (0.92×). Asked to reason first, it names few stars and scores 0.288, below the fully fine-tuned narrator. The larger model is more selective: it names the odd star at about the same rate as the tool relative to the number of stars it names. However, it often gets the odd star's property wrong, and scores 0.341 answering directly and 0.324 after reasoning, no different from the fully fine-tuned narrator (p=0.7 in both cases). In most groups (94%) the larger model was still reasoning when it reached the length limit of 4,096 tokens and was made to answer from what it had, so its reasoning result holds for this budget only. The gap to the LoRA narrators also holds for every group size (Section A.8). Reading the numbers lets a model describe the stars, but not reliably pick out the one that does not belong.

4.6 frontier models on the raw observations

The narrators read features produced by pre-trained encoders. A natural baseline is a frontier general-purpose model given the observations themselves. We show Claude Opus 5.5 and Claude Fable 5.1 each star's raw observations, namely plots of its Kepler and TESS light curves and its LAMOST and APOGEE spectra, together with its Gaia measurements as text, with no derived stellar parameters (Section A.10). The models answer zero-shot, without tools and without any training on this data. Table 9 compares them with the narrators on the same items.

Table 8: **Giving the model the numbers as text**, on the 1,203 test groups with an odd star (all five kinds). The text-reading models see every catalog value plus the binary and open-cluster flags. *score* and *named* are as in Table 7; *stars named* is the fraction of the group the text names; *lift* divides *named* by it (1.0 = no better than naming that many stars at random). Every text-reading model scores below the LoRA narrators ($p < 10^{-14}$, exact McNemar on *score*).

	score	named	lift	stars named
AstroSage-8B, reads the table	0.473	0.761	0.92×	83%
AstroSage-8B, reads the table, reasons first	0.288	0.447	1.82×	25%
Qwen3.5-27B, reads the table	0.341	0.512	3.63×	14%
Qwen3.5-27B, reads the table, reasons first	0.324	0.570	4.07×	14%
narrator, full fine-tune	0.332	0.442	3.36×	13%
narrator, LoRA, FT schedule	0.510	0.672	3.39×	20%
narrator, LoRA	0.633	0.766	**4.31**×	18%
narrator + tool	**0.662**	**0.797**	4.08×	20%

On groups, both frontier models score 0.20, below every narrator ($p < 10^{-13}$ against the LoRA narrators, and $p = 0.005$ against full fine-tuning), and they make about twice as many contradictions. On single stars and pairs, Opus states class labels and comparisons correctly less often than any narrator. The narrators have the advantage of being trained on this task, on descriptions of these same stars (Section 3), so this measures what a general-purpose model does out of the box. Its result does not bound an agent that can run code on the observations.

Table 9: **Frontier models on the raw observations, zero-shot**, paired on the same items. Groups are judge-scored ($n = 177$ with an odd member); single stars ($n = 1{,}463$) and pairs ($n = 1{,}000$) are scored by the verifier as in Table 1. Fable was run on groups only.

	Opus 5.5	Fable 5.1	full fine-tune	LoRA	LoRA + tool
group, score	0.203	0.203	0.333	0.627	**0.644**
group, named	0.339	0.345	0.435	0.734	**0.797**
group, contradictions	3.29	3.14	1.79	**1.36**	1.51
single, numeric acc.	0.394	—	0.667	0.730	**0.796**
single, qualitative acc.	0.636	—	0.892	0.916	**0.935**
pair, numeric acc.	0.534	—	0.774	0.800	**0.853**
pair, relational acc.	0.750	—	0.937	0.949	**0.970**

4.7 Templates vs. LLM prose

Everything above is trained on prose written by the deterministic templater. The obvious alternative is to let a capable LLM write it, which has the potential for better diversity and fluency. To compare the two, we built an LLM-based corpus: the writer (DeepSeek-V4) was given the same catalog facts and asked to write the group descriptions, and every claim it made was checked against the record before the text was accepted. The training text is, in that sense, true. Table 10 compares the two corpora with everything else held fixed: same base model, same prompt, and no tool (both narrators read the data tokens only and are fully fine-tuned). We see that training on the writer's fluent prose makes the narrator invent: contradictions per group go from 1.9 to 7.2, and not one group out of 1,191 is described fully correctly (odd star and property named, typical members described correctly, and no statement contradicting the record). The training text was verified true; the model trained on it is not. This is aligned with the finding of [42] who showed that template-based serialization outperforms LLM-based serialization for tabular data. The two corpora differ in how much they commit to. Per group description, the templater writes a median of 21 words containing 5 numbers and names 17% of the members; it states the population and the member that does not belong, and says nothing about the rest. The writer uses 82 words containing 17 numbers and names 83% of the members, giving most of them a full quantitative profile. Training on the second teaches the narrator to do the same. On the test groups, the narrator trained on writer prose writes a median of 78 words against 17, states 8 values per group against 1, and names 72% of the members against 26%. Most of its extra contradictions (70%) are such numeric claims, and most of those concern members *other*

than the outlier, whose values the template-trained narrator rarely states at all. Per stated value, the two narrators are contradicted at a similar rate (0.52 vs. 0.46): the writer-trained narrator is wrong more often mainly because it says more. Values for the typical members are hard to read for both narrators: for either one about 40% of them contradict the record (verifier). A narrator trained to state them for every member therefore mostly adds errors. This is not the whole story: on the outlier itself, the writer-trained narrator's stated values contradict the record far more often (18% vs. 2% of values, verifier). Verification of the training text does not help in either case, because the problem is not that the text is false; it is that it asserts more than the input determines.

Table 10: **Which prose to train on.** AstroSage-8B, same prompt, latent tokens only (no tool), full fine-tune; greedy, judge-scored, on the $1{,}191$ groups with an odd member judged for both narrators (paired; hence 0.333 here vs. 0.332 on the $1{,}203$ groups of Table 2). *score* is as in Table 2; *covered* = the text correctly describes the typical (non-odd) members; *contradictions/group* = the number of statements the judge marks as contradicting the record; *all correct* = score, covered, and no contradiction.

training prose	score	covered	contradictions/group	all correct
template	**0.333**	**0.573**	**1.88**	**0.128**
writer (DeepSeek-V4), verified	0.258	0.178	7.16	0.000

5 Limitations

Training recipes were not tuned. We compare full fine-tuning and LoRA with the settings each happened to be trained with, and did not search hyperparameters for either. A different recipe might close the gap for full fine-tuning; our claim is that LoRA performs better without such a search, not that full fine-tuning cannot match it. The LoRA narrator trained with the full fine-tuning schedule was stopped before it converged, so the part of the gain we assign to the optimizer schedule (Table 3) also includes the effect of training length.

The tool has catalog labels. The stellar tool models the aligned features jointly with catalog labels and receives those labels at test time, which narrators without the tool do not. The comparison with and without the tool is therefore a comparison with and without auxiliary data, and we do not separate the part of the gain due to the learned density from the part due to the labels. The tool's gain is concentrated on groups whose odd star was never odd in training, where binarity is the odd property in more than half of the groups.

Frontier models were tested zero-shot. The frontier models received the raw observations without tools and without any training on this data, while the narrators read pre-trained features and were trained on descriptions of these stars. The comparison shows what a general-purpose model does out of the box. It does not bound what an agent could do with code execution over the same observations, which we did not complete.

One seed per narrator. Each narrator was trained once. Our claims rest on large paired differences on the same test items and on the same pattern across two datasets.

6 Conclusion

We asked whether a language model can reason directly over learned scientific features, beyond describing them. On single objects and pairs it can, on both datasets. On questions that require reasoning, the main result concerns training rather than architecture. A fully fine-tuned narrator fails to find the odd star in a group, and fails more as the group grows. A narrator trained with LoRA on a frozen base, on the same text and the same features, succeeds in about two thirds of groups and does not lose accuracy with group size, and the same comparison also favours LoRA on galaxies. Neither method was tuned, so a better recipe for full fine-tuning may exist, but LoRA gives this result without one, and most of the gain remains when the two share an optimizer schedule.

Latent tool use adds to this in a specific place. A small flow-matching model that receives catalog labels as auxiliary data, and returns its output as a latent token, improves the identification of outliers that never appeared as outliers in training ($0.62 \rightarrow 0.75$), including when binary stars are excluded ($0.78 \rightarrow 0.88$). It adds little where the training data offers precedent, and nothing on galaxies, where its output repeats what the narrator already reads. Text is a poor substitute: models given the catalog numbers score below the LoRA narrator, and frontier models given the raw observations zero-shot score lower still. Verified templated text remains preferable to fluent LLM-written text as training data.

For a language model reading scientific data, these results suggest first adapting it with a low-rank adapter, and then adding dedicated models with auxiliary data where the task goes beyond what the training data covers. Why the frozen base preserves this ability, and how frontier agents with code execution compare, remain open.

References

[1] Roussel Rahman and Aashwin Ananda Mishra. A Fragile Number Sense: Probing the Elemental Limits of Numerical Reasoning in LLMs. *arXiv e-prints*, art. arXiv:2509.06332, September 2025. doi: 10.48550/arXiv.2509.06332.

[2] Siavash Golkar, Mariel Pettee, Michael Eickenberg, Alberto Bietti, Miles Cranmer, Géraud Krawezik, François Lanusse, Michael McCabe, Ruben Ohana, Liam Holden Parker, Bruno Régaldo-Saint Blancard, Tiberiu Teşileanu, Kyunghyun Cho, and Shirley Ho. xval: A continuous numerical tokenization for scientific language models. 2023. URL https://api.semanticscholar.org/CorpusID:263622222.

[3] Timo Schick, Jane Dwivedi-Yu, Roberto Dessì, Roberta Raileanu, Maria Lomeli, Luke Zettlemoyer, Nicola Cancedda, and Thomas Scialom. Toolformer: Language models can teach themselves to use tools. *ArXiv*, abs/2302.04761, 2023. URL https://api.semanticscholar.org/CorpusID:256697342.

[4] Luyu Gao, Aman Madaan, Shuyan Zhou, Uri Alon, Pengfei Liu, Yiming Yang, Jamie Callan, and Graham Neubig. Pal: Program-aided language models. *ArXiv*, abs/2211.10435, 2022. URL https://api.semanticscholar.org/CorpusID:253708270.

[5] Shibo Hao, Tianyang Liu, Zhen Wang, and Zhiting Hu. Toolkengpt: Augmenting frozen language models with massive tools via tool embeddings. *ArXiv*, abs/2305.11554, 2023. URL https://api.semanticscholar.org/CorpusID:258823133.

[6] Francisco Villaescusa-Navarro, Boris Bolliet, Pablo Villanueva-Domingo, Adrian E. Bayer, Aidan Acquah, Chetana Amancharla, Almog Barzilay-Siegal, Pablo Bermejo, Camille Bilodeau, Pablo Cárdenas Ramírez, Miles Cranmer, Urbano L. França, ChangHoon Hahn, Yan-Fei Jiang, Raul Jimenez, Jun-Young Lee, Antonio Lerario, Osman Mamun, Thomas Meier, Anupam A. Ojha, Pavlos Protopapas, Shimanto Roy, David N. Spergel, Pedro Tarancón-Álvarez, Ujjwal Tiwari, Matteo Viel, Digvijay Wadekar, Chi Wang, Bonny Y. Wang, Licong Xu, Yossi Yovel, Shuwen Yue, Wen-Han Zhou, Qiyao Zhu, Jiajun Zou, and Íñigo Zubeldia. The Denario project: Deep knowledge AI agents for scientific discovery. *arXiv e-prints*, art. arXiv:2510.26887, October 2025. doi: 10.48550/arXiv.2510.26887.

[7] Zhenzhen Zhuang, Jiandong Chen, Hongfeng Xu, Yuwen Jiang, and Jialiang Lin. Large language models for automated scholarly paper review: A survey. *arXiv e-prints*, art. arXiv:2501.10326, January 2025. doi: 10.48550/arXiv.2501.10326.

[8] Kyle Swanson, Wesley Wu, Nash L. Bulaong, John E. Pak, and James Zou. The Virtual Lab of AI agents designs new SARS-CoV-2 nanobodies. *Nature*, 646(8085):716–723, October 2025. ISSN 1476-4687. doi: 10.1038/s41586-025-09442-9. URL https://doi.org/10.1038/s41586-025-09442-9.

[9] Jianwei Yang, Ohm Rishabh Venkatachalam, Mohammad Kianezhad, Sharvaree P. Vadgama, and Rose Yu. Think like a scientist: Physics-guided llm agent for equation discovery. *ArXiv*, abs/2602.12259, 2026. URL https://api.semanticscholar.org/CorpusID:285541022.

[10] Le Zhuo, Zewen Chi, Minghao Xu, Heyan Huang, Heqi Zheng, Conghui He, Xian-Ling Mao, and Wentao Zhang. Protllm: An interleaved protein-language llm with protein-as-word pre-training. In *Annual Meeting of the Association for Computational Linguistics*, 2024. URL https://api.semanticscholar.org/CorpusID:268379189.

[11] Edo Dotan, Iris Lyubman, Marcelo Ehrlich, Eran Bacharach, Tal Pupko, and Yonatan Belinkov. Betadescribe: Providing rich descriptions from protein sequences. *Proceedings of the National Academy of Sciences*, 123(27):e2537345123, 2026. doi: 10.1073/pnas.2537345123. URL https://www.pnas.org/doi/abs/10.1073/pnas.2537345123.

[12] Henry W. Leung and Jo Bovy. Towards an astronomical foundation model for stars with a transformer-based model. *MNRAS*, 527(1):1494–1520, January 2024. doi: 10.1093/mnras/stad3015.

[13] Liam Parker, Francois Lanusse, Jeff Shen, Ollie Liu, Tom Hehir, Leopoldo Sarra, Lucas Meyer, Micah Bowles, Sebastian Wagner-Carena, Helen Qu, Siavash Golkar, Alberto Bietti, Hatim Bourfoune, Nathan Casserau, Pierre Cornette, Keiya Hirashima, Geraud Krawezik, Ruben Ohana, Nicholas Lourie, Michael McCabe, Rudy Morel, Payel Mukhopadhyay, Mariel Pettee, Bruno Regaldo-Saint Blancard, Kyunghyun Cho, Miles Cranmer, and Shirley Ho. AION-1: Omnimodal Foundation Model for Astronomical Sciences. *arXiv e-prints*, art. arXiv:2510.17960, October 2025.

[14] Michael J. Smith, Ryan J. Roberts, Eirini Angeloudi, and Marc Huertas-Company. AstroPT: Scaling Large Observation Models for Astronomy. *arXiv e-prints*, art. arXiv:2405.14930, May 2024. doi: 10.48550/arXiv.2405.14930.

[15] Euclid Collaboration, M. Siudek, M. Huertas-Company, M. Smith, G. Martinez-Solaeche, F. Lanusse, S. Ho, E. Angeloudi, P. A. C. Cunha, H. Domínguez Sánchez, M. Dunn, Y. Fu, P. Iglesias-Navarro, J. Junais, J. H. Knapen, B. Laloux, M. Mezcua, W. Roster, G. Stevens, J. Vega-Ferrero, N. Aghanim, B. Altieri, A. Amara, S. Andreon, N. Auricchio, H. Aussel, C. Baccigalupi, M. Baldi, S. Bardelli, P. Battaglia, A. Biviano, A. Bonchi, E. Branchini, M. Brescia, J. Brinchmann, S. Camera, G. Cañas-Herrera, V. Capobianco, C. Carbone, J. Carretero, S. Casas, F. J. Castander, M. Castellano, G. Castignani, S. Cavuoti, K. C. Chambers, A. Cimatti, C. Colodro-Conde, G. Congedo, C. J. Conselice, L. Conversi, Y. Copin, F. Courbin, H. M. Courtois, M. Cropper, A. Da Silva, H. Degaudenzi, G. De Lucia, A. M. Di Giorgio, J. Dinis, C. Dolding, H. Dole, F. Dubath, C. A. J. Duncan, X. Dupac, S. Dusini, S. Escoffier, M. Farina, R. Farinelli, F. Faustini, S. Ferriol, F. Finelli, S. Fotopoulou, M. Frailis, E. Franceschi, S. Galeotta, K. George, B. Gillis, C. Giocoli, J. Gracia-Carpio, B. R. Granett, A. Grazian, F. Grupp, S. Gwyn, S. V. H. Haugan, W. Holmes, I. M. Hook, F. Hormuth, A. Hornstrup, K. Jahnke, M. Jhabvala, E. Keihänen, S. Kermiche, A. Kiessling, B. Kubik, M. Kümmel, M. Kunz, H. Kurki-Suonio, Q. Le Boulc'h, A. M. C. Le Brun, D. Le Mignant, S. Ligori, P. B. Lilje, V. Lindholm, I. Lloro, G. Mainetti, D. Maino, E. Maiorano, O. Mansutti, S. Marcin, O. Marggraf, M. Martinelli, N. Martinet, F. Marulli, R. Massey, S. Maurogordato, H. J. McCracken, E. Medinaceli, S. Mei, M. Melchior, Y. Mellier, M. Meneghetti, E. Merlin, G. Meylan, A. Mora, M. Moresco, L. Moscardini, R. Nakajima, C. Neissner, S. M. Niemi, J. W. Nightingale, C. Padilla, S. Paltani, F. Pasian, K. Pedersen, W. J. Percival, V. Pettorino, S. Pires, G. Polenta, M. Poncet, L. A. Popa, L. Pozzetti, F. Raison, A. Renzi, J. Rhodes, G. Riccio, E. Romelli, M. Roncarelli, R. Saglia, Z. Sakr, A. G. Sánchez, D. Sapone, B. Sartoris, J. A. Schewtschenko, P. Schneider, T. Schrabback, M. Scodeggio, A. Secroun, G. Seidel, M. Seiffert, S. Serrano, P. Simon, C. Sirignano, G. Sirri, L. Stanco, J. Steinwagner, P. Tallada-Crespí, A. N. Taylor, I. Tereno, S. Toft, R. Toledo-Moreo, F. Torradeflot, I. Tutusaus, L. Valenziano, J. Valiviita, T. Vassallo, G. Verdoes Kleijn, A. Veropalumbo, Y. Wang, J. Weller, A. Zacchei, G. Zamorani, F. M. Zerbi, I. A. Zinchenko, E. Zucca, V. Allevato, M. Ballardini, M. Bolzonella, E. Bozzo, C. Burigana, R. Cabanac, A. Cappi, D. Di Ferdinando, J. A. Escartin Vigo, L. Gabarra, J. Martín-Fleitas, S. Matthew, N. Mauri, R. B. Metcalf, and A. Pezzotta. Euclid Quick Data Release (Q1) Exploring galaxy properties with a multi-modal foundation model. *arXiv e-prints*, art. arXiv:2503.15312, March 2025. doi: 10.48550/arXiv.2503.15312.

[16] Jai Bardhan, Radhikesh Agrawal, Abhiram Tilak, Cyrin Neeraj, and Subhadip Mitra. HEP-JEPA: A foundation model for collider physics using joint embedding predictive architecture. *arXiv e-prints*, art. arXiv:2502.03933, February 2025. doi: 10.48550/arXiv.2502.03933.

[17] Thomas Hayes, Roshan Rao, Halil Akin, Nicholas J. Sofroniew, Deniz Oktay, Zeming Lin, Robert Verkuil, Vincent Q. Tran, Jonathan Deaton, Marius Wiggert, Rohil Badkundri, Irhum Shafkat, Jun Gong, Alexander Derry, Raul S. Molina, Neil Thomas, Yousuf A. Khan, Chetan Mishra, Carolyn Kim, Liam J. Bartie, Matthew Nemeth, Patrick D. Hsu, Tom Sercu, Salvatore Candido, and Alexander Rives. Simulating 500 million years of evolution with a language model. *Science*, 387(6736):850–858, 2025. doi: 10.1126/science.ads0018. URL `https://www.science.org/doi/abs/10.1126/science.ads0018`.

[18] Cristian Bodnar, Wessel P. Bruinsma, Ana Lucic, Megan Stanley, Anna Allen, Johannes Brandstetter, Patrick Garvan, Maik Riechert, Jonathan A. Weyn, Haiyu Dong, Jayesh K. Gupta, Kit Thambiratnam, Alexander T. Archibald, Chun-Chieh Wu, Elizabeth Heider, Max Welling, Richard E. Turner, and Paris Perdikaris. A foundation model for the Earth system. *Nature*, 641

(8065):1180–1187, May 2025. ISSN 1476-4687. doi: 10.1038/s41586-025-09005-y. URL https://doi.org/10.1038/s41586-025-09005-y.

[19] Josh Abramson, Jonas Adler, Jack Dunger, Richard Evans, Tim Green, Alexander Pritzel, Olaf Ronneberger, Lindsay Willmore, Andrew J. Ballard, Joshua Bambrick, Sebastian W. Bodenstein, David A. Evans, Chia-Chun Hung, Michael O'Neill, David Reiman, Kathryn Tunyasuvunakool, Zachary Wu, Akvilė Žemgulytė, Eirini Arvaniti, Charles Beattie, Ottavia Bertolli, Alex Bridgland, Alexey Cherepanov, Miles Congreve, Alexander I. Cowen-Rivers, Andrew Cowie, Michael Figurnov, Fabian B. Fuchs, Hannah Gladman, Rishub Jain, Yousuf A. Khan, Caroline M. R. Low, Kuba Perlin, Anna Potapenko, Pascal Savy, Sukhdeep Singh, Adrian Stecula, Ashok Thillaisundaram, Catherine Tong, Sergei Yakneen, Ellen D. Zhong, Michal Zielinski, Augustin Žídek, Victor Bapst, Pushmeet Kohli, Max Jaderberg, Demis Hassabis, and John M. Jumper. Accurate structure prediction of biomolecular interactions with AlphaFold 3. *Nature*, 630(8016):493–500, 2024. ISSN 1476-4687. doi: 10.1038/s41586-024-07487-w. URL https://doi.org/10.1038/s41586-024-07487-w.

[20] Ming Hu, Chenglong Ma, Wei Li, Wanghan Xu, Jiamin Wu, Jucheng Hu, Tianbin Li, Guohang Zhuang, Jiaqi Liu, Yingzhou Lu, Ying Chen, Chaoyang Zhang, Cheng Tan, Jie Ying, Guocheng Wu, Shujian Gao, Pengcheng Chen, Jiashi Lin, Haitao Wu, Lulu Chen, Fengxiang Wang, Yuanyuan Zhang, Xiangyu Zhao, Feilong Tang, Encheng Su, Junzhi Ning, Xinyao Liu, Ye Du, Changkai Ji, Pengfei Jiang, Cheng Tang, Ziyan Huang, Jiyao Liu, Jiaqi Wei, Yuejin Yang, Xiang Zhang, Guangshuai Wang, Yue Yang, Huihui Xu, Ziyang Chen, Yizhou Wang, Chen Tang, Jianyu Wu, Yuchen Ren, Siyuan Yan, Zhonghua Wang, Zhongxing Xu, Shiyan Su, Shangquan Sun, Runkai Zhao, Zhisheng Zhang, Dingkang Yang, Jinjie Wei, Jiaqi Wang, Jiahao Xu, Jiangtao Yan, Wenhao Tang, Hongze Zhu, Yu Liu, Fudi Wang, Yiqing Shen, Yuanfeng Ji, Yanzhou Su, Tong Xie, Hongming Shan, Chun-Mei Feng, Zhi Hou, Diping Song, Lihao Liu, Yanyan Huang, Lequan Yu, Bin Fu, Shujun Wang, Xiaomeng Li, Xiaowei Hu, Yun Gu, Ben Fei, Benyou Wang, Yuewen Cao, Minjie Shen, Jie Xu, Haodong Duan, Fang Yan, Hongxia Hao, Jielan Li, Jiajun Du, Yanbo Wang, Imran Razzak, Zhongying Deng, Chi Zhang, Lijun Wu, Conghui He, Zhaohui Lu, Jinhai Huang, Wenqi Shao, Yihao Liu, Siqi Luo, Yi Xin, Xiaohong Liu, Fenghua Ling, Yuqiang Li, Aoran Wang, Siqi Sun, Qihao Zheng, Nanqing Dong, Tianfan Fu, Dongzhan Zhou, Yan Lu, Wenlong Zhang, Jin Ye, Jianfei Cai, Yirong Chen, Wanli Ouyang, Yu Qiao, Zongyuan Ge, Shixiang Tang, Junjun He, Chunfeng Song, Lei Bai, and Bowen Zhou. A Survey of Scientific Large Language Models: From Data Foundations to Agent Frontiers. *arXiv e-prints*, art. arXiv:2508.21148, August 2025. doi: 10.48550/arXiv.2508.21148.

[21] Steffen Eger, Yong Cao, Jennifer D'Souza, Andreas Geiger, Christian Greisinger, Stephanie Gross, Yufang Hou, Brigitte Krenn, Anne Lauscher, Yizhi Li, Chenghua Lin, Nafise Sadat Moosavi, Wei Zhao, and Tristan Miller. Transforming Science with Large Language Models: A Survey on AI-assisted Scientific Discovery, Experimentation, Content Generation, and Evaluation. *arXiv e-prints*, art. arXiv:2502.05151, February 2025. doi: 10.48550/arXiv.2502.05151.

[22] Tong Xie, Yuwei Wan, Wei Huang, Zhenyu Yin, Yixuan Liu, Shaozhou Wang, Qingyuan Linghu, Chunyu Kit, Clara Grazian, Wenjie Zhang, Imran Razzak, and Bram Hoex. DARWIN Series: Domain Specific Large Language Models for Natural Science. *arXiv e-prints*, art. arXiv:2308.13565, August 2023. doi: 10.48550/arXiv.2308.13565.

[23] Lei Bai, Zhongrui Cai, Yuhang Cao, Maosong Cao, Weihan Cao, Chiyu Chen, Haojiong Chen, Kai Chen, Pengcheng Chen, Ying Chen, Yongkang Chen, Yu Cheng, Pei Chu, Tao Chu, Erfei Cui, Ganqu Cui, Long Cui, Ziyun Cui, Nianchen Deng, Ning Ding, Nanqing Dong, Peijie Dong, Shihan Dou, Sinan Du, Haodong Duan, Caihua Fan, Ben Gao, Changjiang Gao, Jianfei Gao, Songyang Gao, Yang Gao, Zhangwei Gao, Jiaye Ge, Qiming Ge, Lixin Gu, Yuzhe Gu, Aijia Guo, Qipeng Guo, Xu Guo, Conghui He, Junjun He, Yili Hong, Siyuan Hou, Caiyu Hu, Hanglei Hu, Jucheng Hu, Ming Hu, Zhouqi Hua, Haian Huang, Junhao Huang, Xu Huang, Zixian Huang, Zhe Jiang, Lingkai Kong, Linyang Li, Peiji Li, Pengze Li, Shuaibin Li, Tianbin Li, Wei Li, Yuqiang Li, Dahua Lin, Junyao Lin, Tianyi Lin, Zhishan Lin, Hongwei Liu, Jiangning Liu, Jiyao Liu, Junnan Liu, Kai Liu, Kaiwen Liu, Kuikun Liu, Shichun Liu, Shudong Liu, Wei Liu, Xinyao Liu, Yuhong Liu, Zhan Liu, Yinquan Lu, Haijun Lv, Hongxia Lv, Huijie Lv, Qitan Lv, Ying Lv, Chengqi Lyu, Chenglong Ma, Jianpeng Ma, Ren Ma, Runmin Ma, Runyuan Ma,

Xinzhu Ma, Yichuan Ma, Zihan Ma, Sixuan Mi, Junzhi Ning, Wenchang Ning, Xinle Pang, Jiahui Peng, Runyu Peng, Yu Qiao, Jiantao Qiu, Xiaoye Qu, Yuan Qu, Yuchen Ren, Fukai Shang, Wenqi Shao, Junhao Shen, Shuaike Shen, Chunfeng Song, Demin Song, Diping Song, Chenlin Su, Weijie Su, Weigao Sun, Yu Sun, Qian Tan, Cheng Tang, Huanze Tang, Kexian Tang, Shixiang Tang, Jian Tong, Aoran Wang, Bin Wang, Dong Wang, Lintao Wang, Rui Wang, Weiyun Wang, Wenhai Wang, Jiaqi Wang, Yi Wang, Ziyi Wang, Ling-I Wu, Wen Wu, Yue Wu, Zijian Wu, Linchen Xiao, Shuhao Xing, Chao Xu, Huihui Xu, Jun Xu, Ruiliang Xu, Wanghan Xu, GanLin Yang, Yuming Yang, Haochen Ye, Jin Ye, Shenglong Ye, Jia Yu, Jiashuo Yu, Jing Yu, Fei Yuan, Yuhang Zang, Bo Zhang, Chao Zhang, Chen Zhang, Hongjie Zhang, Jin Zhang, Qiaosheng Zhang, Qiuyinzhe Zhang, Songyang Zhang, Taolin Zhang, Wenlong Zhang, Wenwei Zhang, Yechen Zhang, Ziyang Zhang, Haiteng Zhao, Qian Zhao, Xiangyu Zhao, Xiangyu Zhao, Bowen Zhou, Dongzhan Zhou, Peiheng Zhou, Yuhao Zhou, Yunhua Zhou, Dongsheng Zhu, Lin Zhu, and Yicheng Zou. Intern-S1: A Scientific Multimodal Foundation Model. *arXiv e-prints*, art. arXiv:2508.15763, August 2025. doi: 10.48550/arXiv.2508.15763.

[24] Anoop Cherian, Radu Corcodel, Siddarth Jain, and Diego Romeres. LLMPhy: Complex Physical Reasoning Using Large Language Models and World Models. *arXiv e-prints*, art. arXiv:2411.08027, November 2024. doi: 10.48550/arXiv.2411.08027.

[25] Siddharth Mishra-Sharma, Yiding Song, and Jesse Thaler. PAPERCLIP: Associating Astronomical Observations and Natural Language with Multi-Modal Models. *arXiv e-prints*, art. arXiv:2403.08851, March 2024. doi: 10.48550/arXiv.2403.08851.

[26] Nolan Koblischke, Liam Parker, Francois Lanusse, Irina Espejo Morales, Jo Bovy, and Shirley Ho. Semantic search for 100M+ galaxy images using AI-generated captions. *arXiv e-prints*, art. arXiv:2512.11982, December 2025. doi: 10.48550/arXiv.2512.11982.

[27] Sharaf Zaman, Michael J. Smith, Pranav Khetarpal, Rishabh Chakrabarty, Michele Ginolfi, Marc Huertas-Company, Maja Jabłońska, Sandor Kruk, Matthieu Le Lain, Sergio José Rodríguez Méndez, and Dimitrios Tanoglidis. AstroLLaVA: towards the unification of astronomical data and natural language. *arXiv e-prints*, art. arXiv:2504.08583, April 2025. doi: 10.48550/arXiv.2504.08583.

[28] Tijmen de Haan, Yuan-Sen Ting, Tirthankar Ghosal, Tuan Dung Nguyen, Alberto Accomazzi, Azton Wells, Nesar Ramachandra, Rui Pan, and Zechang Sun. Achieving GPT-4o level performance in astronomy with a specialized 8B-parameter large language model. *Scientific Reports*, 15(1):13751, April 2025. doi: 10.1038/s41598-025-97131-y.

[29] Maria Tsimpoukelli, Jacob Menick, Serkan Cabi, S. M. Ali Eslami, Oriol Vinyals, and Felix Hill. Multimodal few-shot learning with frozen language models. *ArXiv*, abs/2106.13884, 2021. URL https://api.semanticscholar.org/CorpusID:235658331.

[30] Jean-Baptiste Alayrac, Jeff Donahue, Pauline Luc, Antoine Miech, Iain Barr, Yana Hasson, Karel Lenc, Arthur Mensch, Katie Millican, Malcolm Reynolds, Roman Ring, Eliza Rutherford, Serkan Cabi, Tengda Han, Zhitao Gong, Sina Samangooei, Marianne Monteiro, Jacob Menick, Sebastian Borgeaud, Andy Brock, Aida Nematzadeh, Sahand Sharifzadeh, Mikolaj Binkowski, Ricardo Barreira, Oriol Vinyals, Andrew Zisserman, and Karen Simonyan. Flamingo: a visual language model for few-shot learning. *ArXiv*, abs/2204.14198, 2022. URL https://api.semanticscholar.org/CorpusID:248476411.

[31] Junnan Li, Dongxu Li, Silvio Savarese, and Steven C. H. Hoi. Blip-2: Bootstrapping language-image pre-training with frozen image encoders and large language models. *ArXiv*, abs/2301.12597, 2023. URL https://api.semanticscholar.org/CorpusID:256390509.

[32] Haotian Liu, Chunyuan Li, Qingyang Wu, and Yong Jae Lee. Visual instruction tuning. *ArXiv*, abs/2304.08485, 2023. URL https://api.semanticscholar.org/CorpusID:258179774.

[33] Shikun Liu, Linxi (Jim) Fan, Edward Johns, Zhi-Ding Yu, Chaowei Xiao, and Anima Anandkumar. Prismer: A vision-language model with multi-task experts. *Trans. Mach. Learn. Res.*, 2024, 2023. URL https://api.semanticscholar.org/CorpusID:257365896.

[34] Ashutosh Adhikari and Mirella Lapata. Multimodal latent reasoning via predictive embeddings. *ArXiv*, abs/2604.08065, 2026. URL https://api.semanticscholar.org/CorpusID:287255978.

[35] William J. Borucki, David Koch, Gibor Basri, Natalie Batalha, Timothy Brown, Douglas Caldwell, John Caldwell, Jørgen Christensen-Dalsgaard, William D. Cochran, Edna DeVore, Edward W. Dunham, Andrea K. Dupree, Thomas N. Gautier, John C. Geary, Ronald Gilliland, Alan Gould, Steve B. Howell, Jon M. Jenkins, Yoji Kondo, David W. Latham, Geoffrey W. Marcy, Søren Meibom, Hans Kjeldsen, Jack J. Lissauer, David G. Monet, David Morrison, Dimitar Sasselov, Jill Tarter, Alan Boss, Don Brownlee, Toby Owen, Derek Buzasi, David Charbonneau, Laurance Doyle, Jonathan Fortney, Eric B. Ford, Matthew J. Holman, Sara Seager, Jason H. Steffen, William F. Welsh, Jason Rowe, Howard Anderson, Lars Buchhave, David Ciardi, Lucianne Walkowicz, William Sherry, Elliott Horch, Howard Isaacson, Mark E. Everett, Debra Fischer, Guillermo Torres, John Asher Johnson, Michael Endl, Phillip MacQueen, Stephen T. Bryson, Jessie Dotson, Michael Haas, Jeffrey Kolodziejczak, Jeffrey Van Cleve, Hema Chandrasekaran, Joseph D. Twicken, Elisa V. Quintana, Bruce D. Clarke, Christopher Allen, Jie Li, Haley Wu, Peter Tenenbaum, Ekaterina Verner, Frederick Bruhweiler, Jason Barnes, and Andrej Prsa. Kepler Planet-Detection Mission: Introduction and First Results. *Science*, 327(5968):977, February 2010. doi: 10.1126/science.1185402.

[36] George R. Ricker, Joshua N. Winn, Roland Vanderspek, David W. Latham, Gáspár Á. Bakos, Jacob L. Bean, Zachory K. Berta-Thompson, Timothy M. Brown, Lars Buchhave, Nathaniel R. Butler, R. Paul Butler, William J. Chaplin, David Charbonneau, Jørgen Christensen-Dalsgaard, Mark Clampin, Drake Deming, John Doty, Nathan De Lee, Courtney Dressing, Edward W. Dunham, Michael Endl, Francois Fressin, Jian Ge, Thomas Henning, Matthew J. Holman, Andrew W. Howard, Shigeru Ida, Jon M. Jenkins, Garrett Jernigan, John Asher Johnson, Lisa Kaltenegger, Nobuyuki Kawai, Hans Kjeldsen, Gregory Laughlin, Alan M. Levine, Douglas Lin, Jack J. Lissauer, Phillip MacQueen, Geoffrey Marcy, Peter R. McCullough, Timothy D. Morton, Norio Narita, Martin Paegert, Enric Palle, Francesco Pepe, Joshua Pepper, Andreas Quirrenbach, Stephen A. Rinehart, Dimitar Sasselov, Bun'ei Sato, Sara Seager, Alessandro Sozzetti, Keivan G. Stassun, Peter Sullivan, Andrew Szentgyorgyi, Guillermo Torres, Stephane Udry, and Joel Villasenor. Transiting Exoplanet Survey Satellite (TESS). *Journal of Astronomical Telescopes, Instruments, and Systems*, 1:014003, January 2015. doi: 10.1117/1.JATIS.1.1.014003.

[37] Gang Zhao, Yongheng Zhao, Y. Chu, Y. Jing, and L. Deng. Lamost spectral survey — an overview. *Research in Astronomy and Astrophysics*, 2012. doi: 10.1088/1674-4527/12/7/002. URL https://www.semanticscholar.org/paper/6dd33c0f7d83c6cbc8e2ee544cee259518d6f474.

[38] Steven R. Majewski, Ricardo P. Schiavon, Peter M. Frinchaboy, Carlos Allende Prieto, Robert Barkhouser, Dmitry Bizyaev, Basil Blank, Sophia Brunner, Adam Burton, Ricardo Carrera, S. Drew Chojnowski, Kátia Cunha, Courtney Epstein, Greg Fitzgerald, Ana E. García Pérez, Fred R. Hearty, Chuck Henderson, Jon A. Holtzman, Jennifer A. Johnson, Charles R. Lam, James E. Lawler, Paul Maseman, Szabolcs Mészáros, Matthew Nelson, Duy Coung Nguyen, David L. Nidever, Marc Pinsonneault, Matthew Shetrone, Stephen Smee, Verne V. Smith, Todd Stolberg, Michael F. Skrutskie, Eric Walker, John C. Wilson, Gail Zasowski, Friedrich Anders, Sarbani Basu, Stephane Beland, Michael R. Blanton, Jo Bovy, Joel R. Brownstein, Joleen Carlberg, William Chaplin, Cristina Chiappini, Daniel J. Eisenstein, Yvonne Elsworth, Diane Feuillet, Scott W. Fleming, Jessica Galbraith-Frew, Rafael A. García, D. Aníbal García-Hernández, Bruce A. Gillespie, Léo Girardi, James E. Gunn, Sten Hasselquist, Michael R. Hayden, Saskia Hekker, Inese Ivans, Karen Kinemuchi, Mark Klaene, Suvrath Mahadevan, Savita Mathur, Benoît Mosser, Demitri Muna, Jeffrey A. Munn, Robert C. Nichol, Robert W. O'Connell, John K. Parejko, A. C. Robin, Helio Rocha-Pinto, Matthias Schultheis, Aldo M. Serenelli, Neville Shane, Victor Silva Aguirre, Jennifer S. Sobeck, Benjamin Thompson, Nicholas W. Troup, David H. Weinberg, and Olga Zamora. The Apache Point Observatory Galactic Evolution Experiment (APOGEE). *AJ*, 154(3):94, September 2017. doi: 10.3847/1538-3881/aa784d.

[39] Gaia Collaboration, T. Prusti, J. H. J. de Bruijne, A. G. A. Brown, A. Vallenari, C. Babusiaux, C. A. L. Bailer-Jones, U. Bastian, M. Biermann, D. W. Evans, L. Eyer, F. Jansen, C. Jordi,

S. A. Klioner, U. Lammers, L. Lindegren, X. Luri, F. Mignard, D. J. Milligan, C. Panem, V. Poinsignon, D. Pourbaix, S. Randich, G. Sarri, P. Sartoretti, H. I. Siddiqui, C. Soubiran, V. Valette, F. van Leeuwen, N. A. Walton, C. Aerts, F. Arenou, M. Cropper, R. Drimmel, E. Høg, D. Katz, M. G. Lattanzi, W. O'Mullane, E. K. Grebel, A. D. Holland, C. Huc, X. Passot, L. Bramante, C. Cacciari, J. Castañeda, L. Chaoul, N. Cheek, F. De Angeli, C. Fabricius, R. Guerra, J. Hernández, A. Jean-Antoine-Piccolo, E. Masana, R. Messineo, N. Mowlavi, K. Nienartowicz, D. Ordóñez-Blanco, P. Panuzzo, J. Portell, P. J. Richards, M. Riello, G. M. Seabroke, P. Tanga, F. Thévenin, J. Torra, S. G. Els, G. Gracia-Abril, G. Comoretto, M. Garcia-Reinaldos, T. Lock, E. Mercier, M. Altmann, R. Andrae, T. L. Astraatmadja, I. Bellas-Velidis, K. Benson, J. Berthier, R. Blomme, G. Busso, B. Carry, A. Cellino, G. Clementini, S. Cowell, O. Creevey, J. Cuypers, M. Davidson, J. De Ridder, A. de Torres, L. Delchambre, A. Dell'Oro, C. Ducourant, Y. Frémat, M. García-Torres, E. Gosset, J.-L. Halbwachs, N. C. Hambly, D. L. Harrison, M. Hauser, D. Hestroffer, S. T. Hodgkin, H. E. Huckle, A. Hutton, G. Jasniewicz, S. Jordan, M. Kontizas, A. J. Korn, A. C. Lanzafame, M. Manteiga, A. Moitinho, K. Muinonen, J. Osinde, E. Pancino, T. Pauwels, J.-M. Petit, A. Recio-Blanco, A. C. Robin, L. M. Sarro, C. Siopis, M. Smith, K. W. Smith, A. Sozzetti, W. Thuillot, W. van Reeven, Y. Viala, U. Abbas, A. Abreu Aramburu, S. Accart, J. J. Aguado, P. M. Allan, W. Allasia, G. Altavilla, M. A. Álvarez, J. Alves, R. I. Anderson, A. H. Andrei, E. Anglada Varela, E. Antiche, T. Antoja, S. Antón, B. Arcay, A. Atzei, L. Ayache, N. Bach, S. G. Baker, L. Balaguer-Núñez, C. Barache, C. Barata, A. Barbier, F. Barblan, M. Baroni, D. Barrado y Navascués, M. Barros, M. A. Barstow, U. Becciani, M. Bellazzini, G. Bellei, A. Bello García, V. Belokurov, P. Bendjoya, A. Berihuete, L. Bianchi, O. Bienaymé, F. Billebaud, N. Blagorodnova, S. Blanco-Cuaresma, T. Boch, A. Bombrun, R. Borrachero, S. Bouquillon, G. Bourda, H. Bouy, A. Bragaglia, M. A. Breddels, N. Brouillet, T. Brüsemeister, B. Bucciarelli, F. Budnik, P. Burgess, R. Burgon, A. Burlacu, D. Busonero, R. Buzzi, E. Caffau, J. Cambras, H. Campbell, R. Cancelliere, T. Cantat-Gaudin, T. Carlucci, J. M. Carrasco, M. Castellani, P. Charlot, J. Charnas, P. Charvet, F. Chassat, A. Chiavassa, M. Clotet, G. Cocozza, R. S. Collins, P. Collins, and G. Costigan. The Gaia mission. *A&A*, 595:A1, November 2016. doi: 10.1051/0004-6361/201629272.

[40] Caitlin M. Casey, Jeyhan S. Kartaltepe, Nicole E. Drakos, Maximilien Franco, Santosh Harish, Louise Paquereau, Olivier Ilbert, Caitlin Rose, Isabella G. Cox, James W. Nightingale, Brant E. Robertson, John D. Silverman, Anton M. Koekemoer, Richard Massey, Henry Joy McCracken, Jason Rhodes, Hollis B. Akins, Natalie Allen, Aristeidis Amvrosiadis, Rafael C. Arango-Toro, Micaela B. Bagley, Angela Bongiorno, Peter L. Capak, Jaclyn B. Champagne, Nima Chartab, Óscar A. Chávez Ortiz, Katherine Chworowsky, Kevin C. Cooke, Olivia R. Cooper, Behnam Darvish, Xuheng Ding, Andreas L. Faisst, Steven L. Finkelstein, Seiji Fujimoto, Fabrizio Gentile, Steven Gillman, Katriona M. L. Gould, Ghassem Gozaliasl, Christopher C. Hayward, Qiuhan He, Shoubaneh Hemmati, Michaela Hirschmann, Knud Jahnke, Shuowen Jin, Ali Ahmad Khostovan, Vasily Kokorev, Erini Lambrides, Clotilde Laigle, Rebecca L. Larson, Gene C. K. Leung, Daizhong Liu, Tobias Liaudat, Arianna S. Long, Georgios Magdis, Guillaume Mahler, Vincenzo Mainieri, Sinclaire M. Manning, Claudia Maraston, Crystal L. Martin, Jacqueline E. McCleary, Jed McKinney, Conor J. R. McPartland, Bahram Mobasher, Rohan Pattnaik, Alvio Renzini, R. Michael Rich, David B. Sanders, Zahra Sattari, Diana Scognamiglio, Nick Scoville, Kartik Sheth, Marko Shuntov, Martin Sparre, Tomoko L. Suzuki, Margherita Talia, Sune Toft, Benny Trakhtenbrot, C. Megan Urry, Francesco Valentino, Brittany N. Vanderhoof, Eleni Vardoulaki, John R. Weaver, Katherine E. Whitaker, Stephen M. Wilkins, Lilan Yang, and Jorge A. Zavala. COSMOS-Web: An Overview of the JWST Cosmic Origins Survey. *ApJ*, 954(1):31, September 2023. doi: 10.3847/1538-4357/acc2bc.

[41] Alec Radford, Jong Wook Kim, Chris Hallacy, Aditya Ramesh, Gabriel Goh, Sandhini Agarwal, Girish Sastry, Amanda Askell, Pamela Mishkin, Jack Clark, Gretchen Krueger, and Ilya Sutskever. Learning transferable visual models from natural language supervision. *ArXiv*, abs/2103.00020, 2021. URL `https://api.semanticscholar.org/CorpusID:231591445`.

[42] Stefan Hegselmann, Alejandro Buendia, Hunter Lang, Monica Agrawal, Xiaoyi Jiang, and David A. Sontag. Tabllm: Few-shot classification of tabular data with large language models. *ArXiv*, abs/2210.10723, 2022. URL `https://api.semanticscholar.org/CorpusID:252992811`.

A Appendix

A.1 Is the signal in the features? The ranking probe

The group task asks which member does not belong. We test whether that is answerable from the aligned features alone, without a language model in the loop. For each group we rank its members by one quantity and ask where a qualifying star lands. Two readouts are compared on exactly the same groups: the true catalog value, and a cross-fit ridge regression that predicts the same quantity from the aligned feature (5-fold, so no star is scored by a model that saw it). If the probe tracks the catalog, the signal survives the alignment.

Two details matter. First, only the three continuous axes can be ranked by a scalar; binarity and cluster membership are categorical and are excluded (103 of 259 anomalies). Second, the composer designates one intruder but leaves the other members unconstrained on that axis, so a group often contains several stars carrying the same label — 60.5% of anomalies on the full fold, and 81.7% on dwarf/giant. Naming any of them is a true statement about the group, so a readout is credited for the best-placed qualifying star rather than only the designated one. "Extreme" is distance from the group median in units of its median absolute deviation, which does not reveal which direction the answer lies in.

Table 11: **Ranking the odd member from the features alone.** n=156 continuous-axis anomalies over 129 groups. *rank-1* = a qualifying star is the single most extreme member; *top-3* = one is among the three most extreme. The probe reads only what the language model reads.

readout	rank-1	top-3
catalog values (oracle)	0.660	**0.942**
probe, shared subspace (64-d)	0.660	0.840
probe, full feature (1024-d)	**0.686**	0.891
by axis, full feature vs. catalog		
dwarf / giant	0.911 *vs* 0.857	1.000 *vs* 0.929
hot / cool	0.755 *vs* 0.694	0.939 *vs* 0.959
metal rich / poor	0.373 *vs* 0.412	0.725 *vs* 0.941

The probe matches the catalog at rank-1 and keeps 89–95% of it at top-3. The information for this task is therefore in the features.

The catalog row is not a ceiling on the task. It bounds what a *single ranked quantity* can do, and it falls short of 1.0 because the label is a threshold test ("$[\mathrm{Fe/H}] > 0.3$") while a ranking is a distance test: a star can sit far from the group median without crossing the threshold. A readout that models the group jointly, the tool of Table 2, is not bound by it.

Scoring the narrators by the same rule, crediting any equally-qualifying star, raises every narrator ($0.332 \to 0.419$ for the fully fine-tuned narrator, $0.633 \to 0.717$ for the LoRA narrator without the tool, and $0.662 \to 0.756$ for the tool narrator) and leaves the tool narrator's lead over full fine-tuning intact: $+0.33 \to +0.34$ (n=1,203 groups, exact McNemar $p = 7 \times 10^{-81}$).

A.2 The galaxy narrators: token or training?

The galaxy tool narrator differs from the fully fine-tuned narrator in two ways: it reads the tool token, and it is trained with LoRA on a frozen base. To separate the two, we trained a third narrator on the *same* text with the *same* LoRA settings and no tool token, and we also ran the tool narrator with its token scrambled at test time (each galaxy receives the tool token of another test galaxy). All four narrators describe the same $10{,}677$ test galaxies, decoded greedily, and are scored as in Section A.4.

Three things follow. First, the LoRA narrator without the tool token is as good as the one with it: indistinguishable on quiescent galaxies and slightly better on star-forming ones. Second, the fully fine-tuned narrator is clearly worse on quiescent galaxies; most of its errors there come from stating that a quiescent galaxy still forms stars. Third, scrambling the tool narrator's token costs it a little on star-forming galaxies, so the narrator does read the token, but the information it carries is already

Table 12: **Galaxy agreement task, four narrators.** Fraction of test galaxies with the correct verdict. p-values are exact McNemar tests on the same galaxies, against the LoRA narrator with the tool token.

	star-forming (9,609)	quiescent (1,068)
full fine-tune, no tool token	0.980 (p=5×10^{-6})	0.884 (p=2×10^{-23})
LoRA, no tool token	**0.988** (p=0.002)	0.974 (p=0.50)
LoRA + tool token	0.986	**0.978**
LoRA + tool token, scrambled	0.978 (p=3×10^{-9})	0.972 (p=0.34)

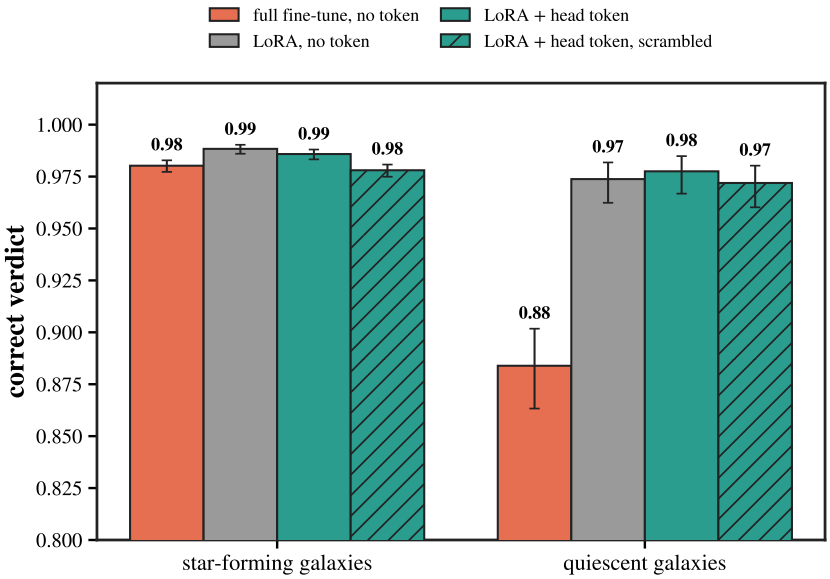


Figure 5: **Galaxy agreement task**, the numbers of Table 12. Error bars are 95% intervals.

available from the data tokens. On galaxies, the improvement over the fully fine-tuned narrator therefore comes from the training, not from the tool token. The stellar equivalent is in Section 4.3.

A.3 The training recipe

Table 13 tests the two input changes of Section 3 on the galaxy data with Llama-3.1-8B. *Novel phrasing* is the fraction of generated descriptions that do not copy a training description (Section A.4); low values mean the narrator recites the templates.

The upper block adds the components one at a time on a 95k-record subset, with everything else fixed. Varying the input format raises accuracy on single objects, but the narrator starts reciting its training text: novel phrasing falls from 0.40 to 0.18 on single objects and from 0.42 to 0.04 on pairs, and pair accuracy drops. Adding the paper sentences reverses this (novel phrasing 0.39 and 0.68) and gives the highest accuracy on single objects and on pair comparisons; only numeric accuracy on pairs stays below the baseline (0.384 vs. 0.416).

The lower block, on the full 288k-record corpus, adds the paper sentences *without* varying the format. On single objects both accuracies rise (numeric 0.540 → 0.647, qualitative 0.836 → 0.870), but copying increases (novel phrasing 0.15 → 0.06). Pairs do improve. This comparison differs from the upper block in corpus size, learning-rate schedule and the share of intervening text taken from papers, so we read it only as consistent with the claim that the paper sentences help when the input format is varied. A one-variable test of the paper sentences without varied formats is in progress.

Table 13: **Training recipe**, galaxies, Llama-3.1-8B, greedy, scored by the verifier. *num.*: numeric accuracy; *qual./rel.*: qualitative (single) or relational (pair) accuracy; *novel*: novel phrasing. Upper block: cosine learning-rate schedule with label smoothing in every row. Lower block: the first two rows use a constant learning rate and no label smoothing; the last row is the full recipe used throughout the paper. Blocks are not comparable with each other.

	single object			pair		
	num.	qual.	novel	num.	rel.	novel
95k-record subset						
baseline	0.500	0.828	0.404	**0.416**	0.731	0.419
+ varied formats	0.519	0.843	0.180	0.340	0.695	0.043
+ varied formats + papers	**0.526**	**0.855**	0.386	0.384	**0.750**	**0.682**
full corpus (288k records)						
baseline	0.540	0.836	0.151	0.374	0.622	0.268
+ papers only	0.647	0.870	0.060	**0.422**	**0.806**	**0.631**
full recipe	**0.750**	**0.881**	**0.537**	0.410	0.795	0.355

A.4 Evaluation

Rule-based verifier (single and pair records). A deterministic program extracts every numeric and categorical claim from the generated text and checks it against the object's catalog record. No model is involved. It reads values written in running text ("a temperature of $5800\,\mathrm{K}$") and in list form ("Surface gravity: 4.3"). A negated statement counts as a claim of the opposite ("not Sun-like" is correct for a star that is not Sun-like), and a galaxy's star-formation trend is checked over the time window the text names (e.g. "declining over the last 5%").

- *Numeric accuracy*: the fraction of stated values within 10% of the catalog value. For quantities given as logarithms ($\log g$, $[\mathrm{Fe/H}]$, $\log M_\star$, $\log \mathrm{sSFR}$) this means 10% in the underlying quantity, i.e. within $\log_{10} 1.1 = 0.041$ dex. The galaxy star-formation trend index is signed and is zero for a flat history, so a relative tolerance is undefined for it; it keeps an absolute tolerance of 0.05. A value off by more than three times the tolerance counts as a *contradiction*. Table 14 gives the accuracy and the mean absolute error of each quantity separately.
- *Qualitative accuracy* (single): the fraction of class words that match the catalog. For stars: evolutionary state (dwarf, subgiant, giant) and group-relative classes (metallicity rank in four classes, hot/cool, fast/slow rotator, young/old, luminous/faint, Sun-like, metal-poor on an absolute scale). For galaxies: morphology class (early/late disk, elliptical, mixed), star-formation state (quiescent, green valley, star-forming, starburst) and mass class.
- *Relational accuracy* (pair): the fraction of comparisons that match the catalog. For stars: directional comparatives (hotter, more massive, larger, older, higher $\log g$, more luminous, more metal-rich, and their opposites). For galaxies: same or different morphology, star-formation-history trend, disturbance and density cluster.
- *Grounded* $= 1-$ the fraction of claims not supported by the record.
- *Novel phrasing* measures whether the narrator writes its own sentences or recites training text. For each generated description we find the most similar description in the training corpus (MinHash nearest neighbour over word 5-grams) and compute the fraction of the generation's 5-word sequences that appear verbatim in it. A generation *copies* a training description if at least half of its 5-word sequences do. Novel phrasing is $1-$ the fraction of generations that copy.

Scoring the galaxy task. We read three statements from each galaxy description: the galaxy's shape (disk or spheroid), whether it still forms stars, and the verdict, i.e., whether the description calls the two in agreement. Each is read with a short list of words. For the verdict, "mismatch", "disagree", "outlier" or "unusual" mark disagreement, and "concordant", "in accord", "concur" or "agree" mark agreement. In the training text these words are exact: every description containing a disagreement word belongs to a disagreeing galaxy, and every one containing an agreement word to an agreeing galaxy (Section A.4). We report the fraction of galaxies with the correct verdict, separately for star-

Table 14: **Numbers stated for single objects**, per quantity, by the Llama-3.1-8B and AstroSage-8B narrators of Table 1 (greedy). *acc.*: fraction within 10% (Section A.4); ***MAE***: mean absolute error in the quantity's unit; n: number of stated values. Quantities stated fewer than 10 times by both models are omitted.

		Llama-3.1-8B			AstroSage-8B		
quantity	unit	n	acc.	MAE	n	acc.	MAE
stars							
$T_{\rm eff}$	K	331	0.99	67	293	1.00	67
$\log g$	dex	355	0.59	0.072	293	0.59	0.074
mass	$M_\odot$	342	0.85	0.050	222	0.83	0.064
radius	$R_\odot$	155	0.57	0.34	180	0.61	0.40
age	Gyr	284	0.34	1.46	178	0.31	1.50
parallax	mas	270	0.43	0.22	147	0.36	0.30
galaxies							
$\log M_\star$	dex	88	0.20	0.19	226	0.23	0.14
$\log \mathrm{sSFR}$	dex	229	0.30	0.12	195	0.27	0.14
redshift	—	739	0.85	0.066	543	0.86	0.045
Sérsic index	—	0	—	—	61	0.16	1.40
formation age	Gyr	82	0.55	0.054	324	0.58	0.17
SFH mean time	—	1692	0.89	0.012	2248	0.87	0.013
SFH trend index	—	1726	0.70	0.048	1890	0.66	0.051
recent mass fraction	—	419	0.65	0.026	1058	0.64	0.023

forming and quiescent galaxies. The split matters because 90% of the test galaxies are star-forming, and the errors concentrate in the quiescent minority, where a narrator must notice that a galaxy has stopped forming stars. Whenever a narrator's verdict is right, its stated shape and star-formation state are also right in more than 99.9% of cases.

The judge (stellar population records). The verifier's extractors expect the template's phrasing, so the population task is also scored by the judge (Qwen3.5-27B), a text-only model that never sees any token. It reads each group description against the ground-truth record and is instructed to judge factual content only, not style. It returns: whether the text names the true odd member (*named*); whether it attributes it to the correct property (*property*); whether the population statement describes the typical members (*covered*); and a list of claims that contradict the record (*contradictions*). We derive *score* = named *and* property, and *all correct* = score, covered, and no contradiction. Only groups that contain an odd member are scored. All narrators are judged on the same groups, so comparisons are paired (exact McNemar for yes/no outcomes, Wilcoxon for contradiction counts). The judge is given no numeric tolerance and is stricter than the verifier: of the numeric contradictions it lists that we could parse, 56–60% (depending on the narrator) are within the verifier's tolerance. Its contradiction counts are therefore an upper bound, and they inflate more for a narrator that states more numbers.

Galaxy agreement task. Each description is reduced to three statements, read with fixed word lists taken from the nine single-galaxy templates of the training text. *Shape*: "late-type disk" or "early-type disk" (disk), "elliptical" or "lenticular" (spheroid); general phrases that state the rule, such as "spheroidal systems", are ignored. *Star formation*: the first of "star-forming" or "still forming stars" (star-forming) and "quiescent" or "has quenched" (quiescent); the galaxy's own state always precedes the rule clause, as in "This disk is quiescent — … disks usually remain star-forming". *Verdict*: the words in Table 15; a description containing words from both lists, or from neither, has no verdict. Table 15 checks every word against the ground truth of the $36{,}654$ single-galaxy training descriptions; the shape and star-formation rules are likewise correct on 100% of them. A galaxy's verdict is scored correct if it matches the catalog. No narrator left a galaxy without a verdict.

A.5 Accuracy per property

Figure 6 breaks the single-object accuracy of AstroSage-8B down by property, on the whole test set ($19{,}010$ stars, $3{,}000$ galaxies). Most class statements are right more than 90% of the time. The exceptions are informative. For stars, the metallicity rank is right in 68% of cases, and most errors

Table 15: **Verdict words in the galaxy training text.** For each word, the number of the 36,654 single-galaxy training descriptions that contain it, and the fraction of those that describe a galaxy whose shape and star-formation history truly disagree.

	word	descriptions	galaxy disagrees
disagreement	mismatch	4,012	100%
	disagree	3,961	100%
	outlier	4,123	100%
	unusual(ly)	6,893	100%
agreement	concordant	8,177	0%
	in accord	8,054	0%
	concur	8,327	0%
	agree	8,177	0%

confuse two neighbouring classes (below and above the median); subgiants are often called dwarfs (59% correct, against 98% for dwarfs and giants). Among stated numbers, temperature and mass are usually within 10%, while radius, surface gravity, parallax and age often are not; age is off by 1.5 Gyr on average. For galaxies, redshift and the mean star-formation time are accurate, while stellar mass, specific star-formation rate and the Sérsic index are rarely within 10% (stellar mass is off by 0.14 dex on average).

A.6 The stellar group task, by kind of odd star

Groups are built by choosing members that share one property and adding an odd star that differs on another, but the other members are not checked on that second property, so some groups contain more than one star with the odd property. Table 16 gives the score for each kind of odd star on all test groups, and on the 556 groups in which the odd star is the only member with its odd property. The tool narrator improves on the fully fine-tuned narrator in every row; most of that gain is already present in the LoRA narrator without the tool, and the tool adds most on binary stars. On the second set the fully fine-tuned narrator again declines with group size (0.44, 0.33 and 0.14 for 4–7, 8–15 and 16–32 stars), while the tool narrator does not (0.70, 0.74, 0.75).

Table 16: **The stellar group task, by kind of odd star**, judge-scored, greedy; score as in Table 2. Rows use the groups whose odd stars all differ on the same property (775 of 1,203; the last row uses all groups). *Unique*: groups in which the odd star is the only member with its odd property. Binarity and open-cluster membership are also given to the tool as catalog flags (Section 4.5). All differences between the fully fine-tuned narrator and the tool narrator are significant (exact McNemar, $p < 10^{-3}$) except giants in the unique groups ($p = 0.33$, $n = 60$).

	all groups				unique			
odd star	n	full FT	LoRA	+ tool	n	full FT	LoRA	+ tool
open-cluster member	120	0.150	0.692	**0.708**	117	0.145	0.684	**0.709**
binary	140	0.157	0.429	**0.550**	47	0.255	0.468	**0.660**
metal-rich	169	0.533	0.799	**0.840**	147	0.551	0.782	**0.850**
hot	161	0.342	0.528	**0.627**	99	0.394	0.606	**0.657**
giant	185	0.227	0.427	**0.449**	60	0.300	**0.417**	0.400
all kinds	1,203	0.332	0.633	**0.662**	556	0.369	0.664	**0.714**

A.7 Does the narrator say what the tokens contain?

To see whether the narrator's errors come from the tokens or from the narrator, we fit two probes for every property on the same inputs the narrator receives (the aligned star tokens; the image and star-formation-history embeddings of a galaxy). The first probe is linear (ridge or logistic regression); the second is non-linear with one hidden layer of 512 units. Both are trained on the training fold and tested on the test fold. For each property we compare them with the narrator on the objects where the narrator stated that property (Figure 7).

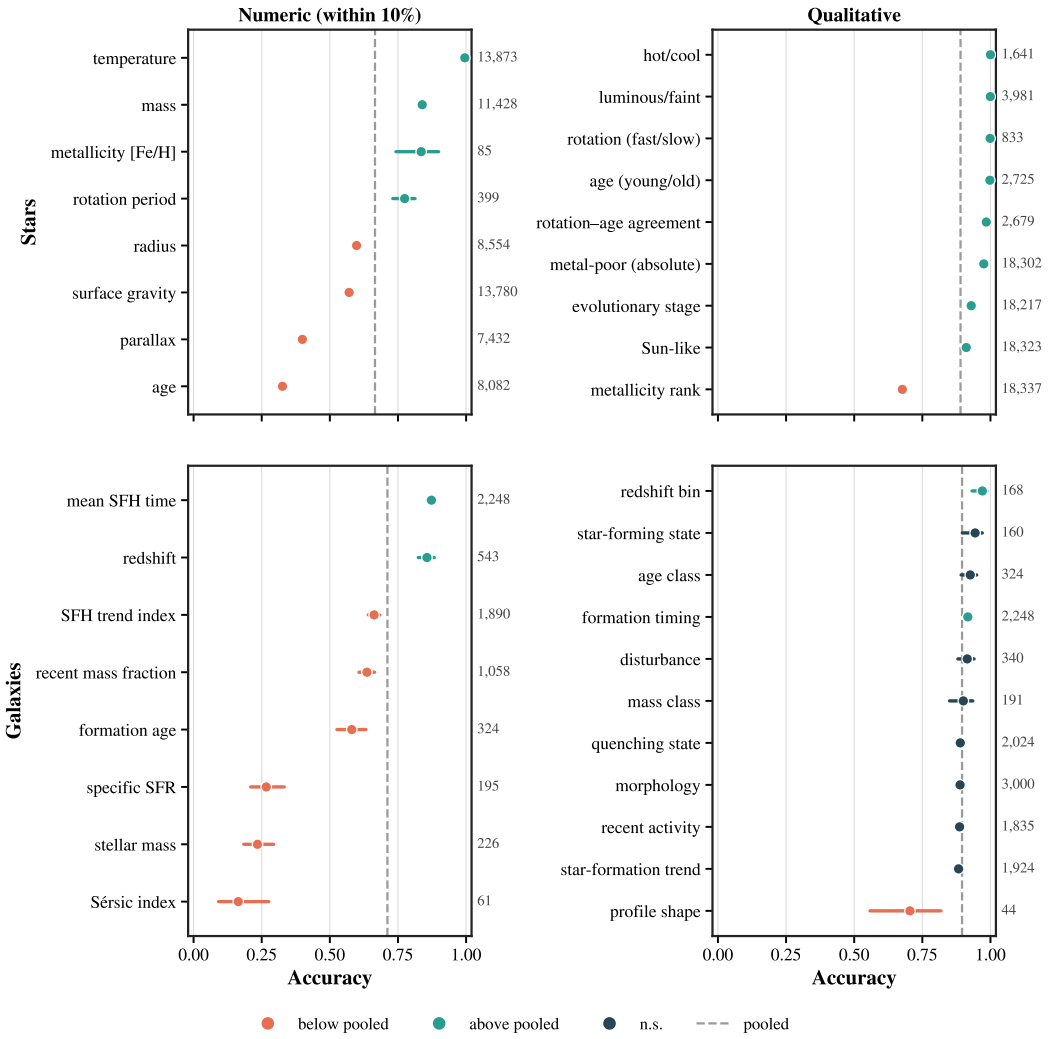


Figure 6: **Single-object accuracy per property**, AstroSage-8B, greedy, whole test set. Left: fraction of stated numbers within 10% of the catalog value (Section A.4); right: fraction of class statements that match the catalog. Bars are 95% intervals; the dashed line is the pooled accuracy; colour marks properties significantly below or above it; numbers at the right are how many statements were scored. Properties stated fewer than 30 times are omitted.

Galaxies. No test galaxy appears in the training text, so this is a clean comparison. The narrator says about as much as the non-linear probe can read. On classes they are equal for morphology, disturbance and the star-formation trend; elsewhere the probe is ahead by 0.03–0.07 (by 0.09 for the Sérsic shape, which the narrator states for 1% of galaxies). On numbers the probe is more accurate for the star-formation-history quantities and the specific star-formation rate, and the narrator is more accurate for redshift and stellar mass. The largest gap is not accuracy but coverage: the narrator states the stellar mass, specific star-formation rate, redshift or Sérsic index for only 2–18% of galaxies, although the probes read them well ($R^2 = 0.85$–0.99, except the Sérsic index, $R^2 = 0.43$).

Stars. The narrator is more accurate than both probes on most numbers, for example radius (0.40 against 0.82 $R_\odot$ for the non-linear probe). Part of this is recall. The narrator states the exact catalog value far more often than a probe can: temperature to the kelvin in 35% of its statements, against 0.5% for the non-linear probe. Almost every such value is written in some training description of that star, which exists for 99.2% of test stars (Section 3); when the value appears in none of them,

exact matches drop to the probe's level. The effect on the average error is small: it is only slightly lower where the training text gave the number. On classes the narrator and the probes agree to within 0.02.

Weak in both. Some properties are hard for the probes too: star age ($R^2 = 0.55$), the metallicity rank (66%, against 25% by chance), and the galaxy Sérsic index. For these, the information is mostly missing from the tokens, and the narrator's errors (Figure 6) reflect the tokens rather than the narrator.

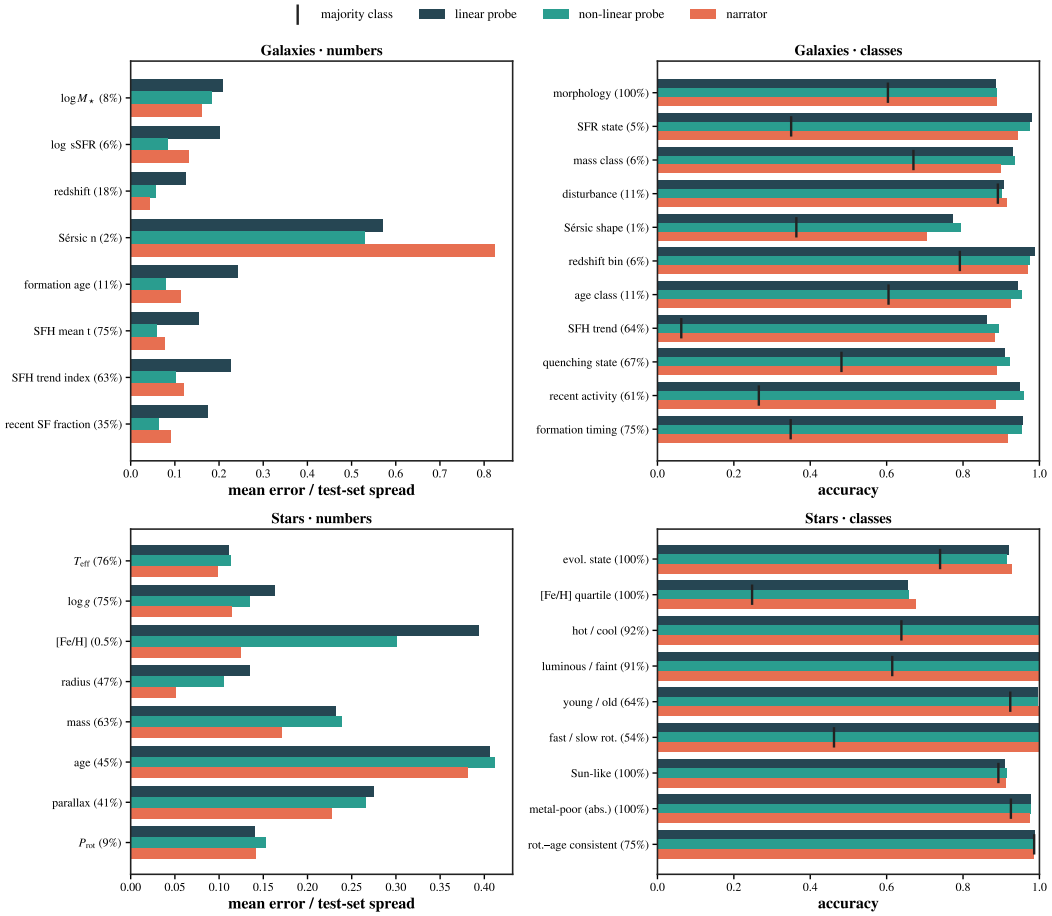


Figure 7: **Probes against the narrator, per property**, AstroSage-8B, single objects. Probes read the same tokens as the narrator. Left: mean absolute error divided by the spread (standard deviation) of the property in the test set; right: accuracy of class statements, with the rate of the most common class (black tick). All bars are on the objects where the narrator stated the property; the percentage next to each name is how often it did. Top: galaxies (3,000, none in the training text); bottom: stars (19,010, 99.2% described in the training text).

A.8 Reading the numbers, by group size

Table 17 breaks Table 8 down by the number of stars in the group. The fully fine-tuned narrator falls from 0.45 to 0.06 as groups grow; the models that read the table fall less, but stay below the LoRA narrators at every size.

A.9 Training recipes of the stellar narrators

Full fine-tuning updates all weights of the language model, with a learning rate of 5×10^{-5}, a cosine schedule with warm-up, and label smoothing of 0.05. The LoRA narrators freeze the base model

Table 17: **Score by number of stars in the group**, for the models of Table 8 (same 1,203 groups; n = groups of that size).

stars in the group	4	6	8	12	16	24	32
n	276	242	191	161	127	112	94
AstroSage-8B, reads the table	0.55	0.52	0.48	0.30	0.50	0.41	0.45
AstroSage-8B, reads the table, reasons first	0.29	0.29	0.33	0.26	0.28	0.33	0.22
Qwen3.5-27B, reads the table	0.40	0.39	0.31	0.37	0.26	0.24	0.30
Qwen3.5-27B, reads the table, reasons first	0.30	0.38	0.37	0.35	0.31	0.22	0.27
narrator, full fine-tune	0.45	0.37	0.44	0.30	0.25	0.13	0.06
narrator, LoRA, FT schedule	0.51	0.52	0.61	0.52	0.50	0.46	0.34
narrator, LoRA	0.60	0.61	0.70	0.70	0.67	0.60	0.53
narrator + tool	**0.68**	**0.63**	**0.71**	**0.70**	**0.68**	**0.62**	**0.56**

and train low-rank adapters (rank 16, $\alpha = 32$) on the attention query and value projections. The tool narrator and the LoRA narrator without the tool use a constant learning rate of 2.83×10^{-4} and no label smoothing. The LoRA narrator with the full fine-tuning schedule uses the learning rate, schedule and label smoothing of the fully fine-tuned narrator; it was stopped after 20 epochs of a 50-epoch schedule, at which point its validation loss was still improving. No hyperparameters were searched for any of these narrators.

A.10 Frontier-model inputs

Each star is shown as fixed plots of its full Kepler light curve, a 20-day segment of it, one TESS sector, and its LAMOST and APOGEE spectra, together with its Gaia measurements as text. No derived stellar parameter is shown. A group shows these panels for every member.

Plots may suit a vision model less than numbers do. We therefore replaced each plot with a table of the same observations, binned to a few hundred values per curve or spectrum, and reran Claude Opus 5.5 on a subset of the same items (Table 18). The results do not change: the model states about the same fraction of numbers and classes correctly, and it finds the odd star in about as few groups. The low score therefore does not come from the plot format.

We also started a third version, in which the model can run code on the unbinned observations. It ran on only three groups before we stopped it, which is too few to report.

Table 18: **Plots against tables as input**, Claude Opus 5.5, zero-shot, paired on the same items. Single stars are scored by the verifier as in Table 9; groups are judge-scored, and we report counts because there are only 16 groups with an odd member.

	plots	tables
single stars ($n = 100$)		
numeric acc.	0.357	0.385
qualitative acc.	0.580	0.590
groups with an odd member ($n = 16$)		
odd star and property named	3	2
odd star named	4	2
contradictions per group	3.06	3.19